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Srs2 binding to PCNA and its sumoylation contribute to RPA antagonism during the DNA damage response

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Abstract

Activation of the DNA damage checkpoint upon genotoxin treatment induces a multitude of cellular changes, such as cell cycle arrest, to cope with genome stress. After prolonged genotoxin treatment, the checkpoint can be downregulated to allow cell cycle and growth resumption. In yeast, downregulation of the DNA damage checkpoint requires the Srs2 DNA helicase, which removes the ssDNA binding complex RPA and the associated Mec1 checkpoint kinase from DNA, thus dampening Mec1 activation. However, it is unclear whether the 'anti-checkpoint' role of Srs2 is temporally and spatially regulated to both allow timely checkpoint termination and to prevent superfluous RPA removal. Here we address this question by examining regulatory elements of Srs2, including its phosphorylation, sumoylation, and protein-interaction sites. Our genetic analyses and checkpoint level assessment suggest that the RPA countering role of Srs2 is promoted by Srs2 binding to PCNA, which is known to recruit Srs2 to subsets of ssDNA regions. RPA antagonism is further fostered by Srs2 sumoylation, which we found depends on the Srs2-PCNA interaction. Srs2 sumoylation is additionally reliant on Mec1 and peaks after Mec1 activity reaches maximal levels. Collectively, our data provide evidence for a two-step model wherein checkpoint downregulation is facilitated by PCNA-mediated Srs2 recruitment to ssDNA-RPA filaments and the subsequent Srs2 sumoylation stimulated upon Mec1 hyperactivation. We propose that this mechanism allows Mec1 hyperactivation to trigger checkpoint recovery.

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eLife assessment

This manuscript reports **valuable** findings on the role of the Srs2 protein in turning off the DNA damage signaling response initiated by Mec1 (human ATR) kinase. The data provide **solid** evidence that Srs2 interaction with PCNA and ensuing SUMO modification is required for checkpoint downregulation. However, experimental evidence with regard to the model that Srs2 acts at gaps after camptothecin-induced DNA damage is currently lacking. The work will be of interest to cell biologists studying genome integrity but would be strengthened by considering the possible role of Rad51 and its removal.

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Introduction

The DNA damage response promotes organismal survival of genotoxic stress by eliciting many cellular changes. This includes a critical signaling process termed the DNA damage checkpoint (DDC)(Enoch, Carr, & Nurse, 1992; Hartwell & Weinert, 1989; Lanz, Dibitetto, & Smolka, 2019). Upon encountering stress, the apical DDC kinases are recruited to DNA lesion sites by the damage sensing proteins and are then activated by activator proteins(Usui, Ogawa, & Petrini, 2001; L. Zou & S. J. Elledge, 2003). A universal DDC sensor is the RPA complex that binds with high affinity and specificity to single-strand DNA (ssDNA), whose abundance rises under almost all types of genotoxic stress(Maréchal & Zou, 2015). Using the budding yeast as an example, RPA bound to the ssDNA filament can interact with Ddc2, the co-factor of the Mec1 checkpoint kinase, and recruit the Mec1-Ddc2 complex to DNA lesion sites(Elledge, 1996; Foiani et al., 2000). Subsequently, Mec1 is activated and can phosphorylate multiple substrates(Harrison & Haber, 2006). Among these is a key downstream effector kinase Rad53, the phosphorylation of which turns on its own kinase activity and yields further phosphorylation. Collectively, the DDC kinases can modify hundreds of proteins, inducing multitude of pathway changes (Lanz, Dibitetto, & Smolka, 2019). For example, DDC can arrest the cell cycle progression to allow more time for genome repair, which is required for cellular survival (Sanchez et al., 1999; Wang et al., 2001).

While DDC activation is crucial for the cell to cope with genotoxic stress, its timely downregulation is equally vital (Waterman, Haber, & Smolka, 2020). Many changes induced by the DDC kinases need to be reversed to allow continued cell growth and recovery. For example, the DDC-mediated cell cycle arrest has to be lifted to allow proliferation and alternative DNA repair permitted at other cell cycle phases (Pellicioli et al., 2001; Waterman, Haber, & Smolka, 2020). Thus far, DDC downregulation has been mainly elucidated using the model organism yeast, with several strategies being identified (Waterman, Haber, & Smolka, 2020). One of these is mediated by the DNA helicase Srs2 that targets the most upstream step of the DDC pathway through removing RPA and the associated Mec1-Ddc2 from chromatin (Dhingra et al., 2021). The DDC dampening (anti-checkpoint) function of Srs2 was shown to be separable from its roles in removing the recombinase Rad51 from ssDNA (anti-recombination) during homologous recombination (Dhingra et al., 2021). Considering that other organisms also possess DNA helicases capable of removing RPA from ssDNA (Hormeno et al., 2022; Jenkins et al., 2021; Shorrocks et al., 2021), it is plausible that an anti-checkpoint strategy involving RPA antagonism may be present beyond yeast.

Currently, little is known about how checkpoint dampening can be temporally controlled to minimize interference with the initial DDC activation that must occur in the early phase of the DNA damage response. In the case of the Srs2-mediated checkpoint regulation, it is also unclear how the process can minimize superfluous removal of RPA from sites of other RPA-mediated processes, such as at ongoing replication forks and transcription bubbles. In addressing these questions, we hypothesize that Srs2-based RPA antagonism should be selective in location and timing, and that the regulatory elements within the Srs2 protein may contribute to the selectivity.

To test the above hypothesis and to define Srs2 features relevant to its RPA antagonism during DDC regulation, we examined six Srs2 protein sites involved in its post-translational modifications and interactions with other proteins. Our genetic analyses of mutations affecting each of these sites suggest that while most of the examined features do not affect RPA antagonism, Srs2 binding to PCNA, which marks the 3' junction of ssDNA gaps, and Srs2 sumoylation contribute to RPA regulation in the context of DDC. Srs2 binding to PCNA was known to recruit Srs2 to replication perturbation sites, while the roles of Srs2 sumoylation were less studied but can affect rDNA recombination (Kolesar et al., 2012; Moldovan, Pfander, & Jentsch, 2007; Papouli et al., 2005; Pfander et al., 2005; Saponaro et al., 2010). Unexpectedly, we found that Srs2 PCNA binding and its sumoylation are functionally related, as Srs2 binding to PCNA promotes its sumoylation. Moreover, Srs2 sumoylation level peaks late in the response to genomic stress after the DDC kinases are hyper-activated, and this modification depends on Mec1. Collectively, our data support a model wherein Srs2 recruitment to DNA lesion sites via binding to PCNA and subsequent sumoylation in a Mec1-dependent manner help to counter RPA functions and the Mec1 checkpoint.

Results

Srs2 regulatory features and their disabling mutants

To understand which Srs2 attributes are relevant to RPA regulation during the DNA damage response, we examined mutant alleles that perturb known features of the Srs2 protein. Srs2 was reported to contain a helicase domain at its N-terminal region and three protein-protein interaction sites at its C-terminal region (**Fig. 1A**) (Niu & Klein, 2017). We have previously shown that the Srs2 ATPase-dead allele (*srs2-K41A*) fails to downregulate RPA and Mec1-mediated DDC, indicating a requirement of ATPase-driven stripping of RPA from ssDNA (Dhingra et al., 2021) . The reported Srs2 protein-bindings sites include a Rad51 binding domain (Rad51-BD), a PCNA interaction motif (PIM), and a SUMO interaction motif (SIM) (**Fig. 1A**) (Colavito et al., 2009; Kolesar et al., 2016; Kolesar et al., 2012). We used CRISPR-based editing to generate the *srs2-ΔPIM* (Δ1159-*11*63 amino acids) and the *srs2-SIMmut* ($_{1170}$ IIVID $_{1173}$ to $_{1170}$ AAAAD $_{1173}$) alleles based on previous reports and examined an established *srs2-ΔRad51BD* allele (Δ875-902 amino acids) (Colavito et al., 2009; Kolesar et al., 2016; Kolesar et al., 2012). We note that despite Srs2 removal of RPA from ssDNA, an interaction between the two proteins has yet to be reported.

We also queried post-translational modifications reported for Srs2 by mutating the corresponding modification sites (**Fig. 1A**). First, seven Cdk1 phosphorylation consensus sites of Srs2 were mutated to produce the *srs2-7AV* (T604V, S698A, S879A, S893A, S938A, S950A, and S965A) allele (Chiolo et al., 2005), which was previously shown to affect recombinational repair. Second, three sumoylation sites of Srs2 were mutated to generate *srs2-3KR* (K1081R, K1089R, and K1142R) (Saponaro et al., 2010). Finally, we mutated two serine residues (S890 and S933) to alanine to generate *srs2-2SA*, since phosphorylation at these sites was shown in proteomic studies as dependent on Mec1 and genotoxic treatments (Faca et al., 2020). Because the Srs2 C-terminal end contributes to protein functions (Antony et al., 2009; Colavito et al., 2009), the six mutant

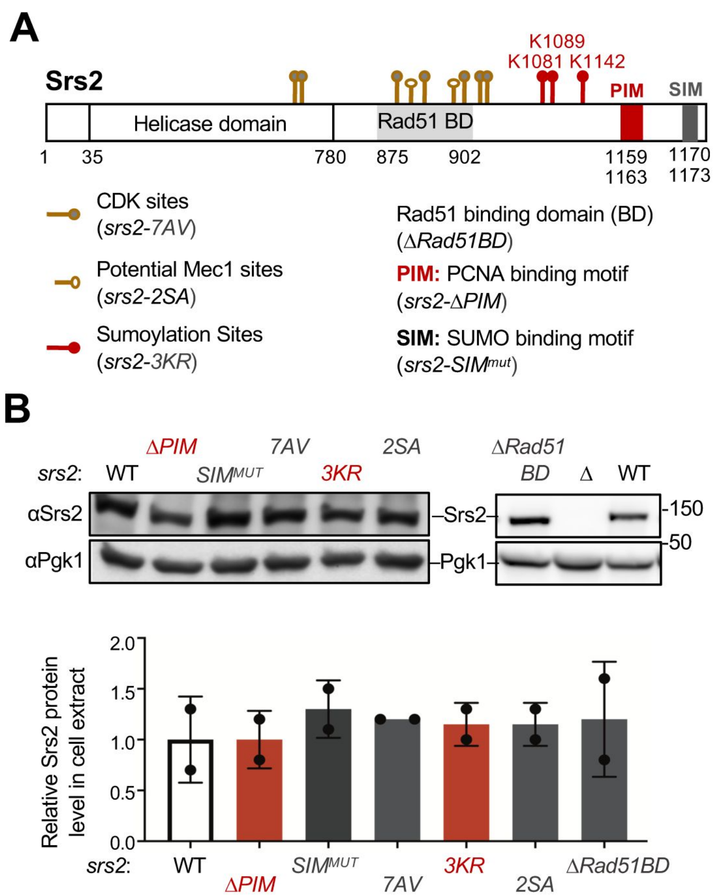

Figure 1.

Regulatory features of Srs2 and corresponding mutants.

(A) Schematic of Srs2 protein domains, regulatory features, and mutations affecting each of its features. Cdk1 phosphorylation sites depicted here include T604, S698, S879, S893, S938, S950, and S965 (Chiolo et al., 2005). Potential Mec1 phosphorylation sites include S890 and S933 (Albuquerque et al., 2015; Faca et al., 2020). Sumoylation sites mapped previously include K1081, K1089, and K1142 (Saponaro et al., 2010). Protein-interaction domains or motifs include the Rad51 binding domain (BD), PCNA Interaction Motif (PIM), and SUMO Interaction Motif (SIM) (Colavito et al., 2009; Kolesar et al., 2016; Kolesar et al., 2012). The residues for these domains are indicated. Mutant alleles disabling each of these features are included in the parentheses.

(B) Mutations affecting Srs2 features do not affect Srs2 protein level. Protein extracts were prepared from asynchronous cells. Srs2 was examined using anti-Srs2 antibody by immunoblotting. Srs2 levels in each mutant were first normalized to the Pgk1 loading control and then to those in wild-type cells. Mean of two biological isolates per genotype is graphed with error bars representing standard deviation (SD).

alleles described above were not tagged with epitopes. Using an anti-Srs2 specific antibody, we showed that all six alleles supported wild-type levels of Srs2 protein based on immunoblotting (**Fig. 1B**).

Mutating PCNA binding and sumoylation sites of Srs2 rescues *rfa1-zm2* CPT sensitivity

Srs2 loss leads to persistent DDC and perturbed recombination, both of which contribute to *srs2Δ* sensitivity to genotoxins, including camptothecin (CPT) that traps Top1 on DNA and methyl methanesulfonate (MMS) that methylates DNA (**Fig. 2A**) (Niu & Klein, 2017). Significantly, our recent study has shown that RPA mutants that reduce its ssDNA binding affinity can rescue the persistent DDC levels in *srs2Δ* cells but not their recombination defects (Dhingra et al., 2021). *rfa1-zm2* (N492D, K493R, K494R) is one of the RPA alleles examined that contains mutations at three DNA contacting residues in the Rfa1 subunit of RPA (Fan & Pavletich, 2012). An RPA variant complex with these mutations exhibits a two-fold increase in dissociation constant compared to wild type RPA (Dhingra et al., 2021). In cells, *rfa1-zm2* supports normal Rfa1 protein level, cell growth, DNA replication, and recombinational repair (Dhingra et al., 2021). However, *rfa1-zm2* and *srs2Δ* are mutually suppressive for genotoxic sensitivities and DDC abnormalities (**Fig. 2B**, S1A) (Dhingra et al., 2021). These data suggest that the *srs2* and *rfa1-zm2* genetic relationship can be used as a readout for the antagonism between Srs2 and RPA during the DNA damage response. Based on this rationale, we employed the genetic suppression as an initial readout in querying whether any of the six *srs2* mutant alleles described above are related to RPA regulation. Given that Srs2's role in DDC dampening was best examined in CPT (Dhingra et al., 2021), experiments were mostly carried out in this condition.

We found that on their own, *srs2-SIM*^{mut}, *-ΔPIM*, and *-7AV* showed different levels of sensitivity toward CPT at a concentration ranging from 2 to 8 µg/ml, while *srs2-ΔRad51BD* and *-3KR* exhibited close to wild-type level of CPT resistance, which are consistent with previous reports (Colavito et al., 2009; Saponaro et al., 2010) (Fig. S1A). In addition, *srs2-2SA,* which has not been examined previously, did not show CPT sensitivity (Fig. S1A). Since *rfa1-zm2* cells exhibited stronger CPT sensitivity than the tested *srs2* alleles (Fig. S1A), we asked if the latter could suppress *rfa1-zm2* CPT sensitivity. We found that both *srs2-ΔPIM* and *-3KR* improved the CPT resistance of *rfa1-zm2* cells on media containing 2 µg/ml CPT (**Fig. 2C**). Rescue was also seen at higher CPT concentrations, with *srs2-ΔPIM* exhibiting a stronger effect than *srs2-3KR* (Fig. S1B). In contrast to *srs2-ΔPIM* and *srs2-3KR*, *srs2-ΔRad51BD* and -2SA showed no effect on *rfa1-zm2* growth on media containing CPT, whereas *srs2*-7AV *and -SIM*^{mut} worsened it (**Fig. 2D**). The differential effects seen for the *srs2* alleles are consistent with the notion that Srs2 has multiple roles in cellular survival of genotoxic stress (Bronstein et al., 2018; Niu & Klein, 2017). The unique suppression seen for mutants affecting Srs2 binding to PCNA and its sumoylation suggests that these features are relevant to RPA regulation during the DNA damage response.

Additional suppressive effects conferred by *srs2-ΔPIM* and *-3KR* toward *rfa1* mutants

We next tested whether the observed *srs2-ΔPIM* and *-3KR* suppression was specific towards *rfa1-zm2* or applicable to other *rfa1* mutants that are also defective in ssDNA binding. While *rfa1-zm2* affects the ssDNA binding domain C (DBD-C) of Rfa1, a widely used *rfa1* allele, *rfa1-t33*, impairs the DBD-B domain (**Fig. 2E**) (Umezu et al., 1998). Unlike *rfa1-zm2, rfa1-t33* additionally decreases Rfa1 protein level and interferes with DNA replication and repair (Deng et al., 2014; Umezu et al., 1998). Despite having these additional defects, CPT survival of *rfa1-t33* cells was also improved by *srs2-ΔPIM* or *srs2-3KR* (**Fig. 2E**). As seen in the case of *rfa1-zm2*, *srs2-ΔPIM* conferred better rescue than *srs2-3KR* toward *rfa1-t33* (**Fig. 2E**). When comparing the two *rfa1* alleles, *srs2* rescue was stronger toward *rfa1-zm2*, likely because *rfa1-t33* has other defects besides reduced ssDNA binding (**Fig. 2C**, 2E). In either case, *srs2* mutants only conferred partial

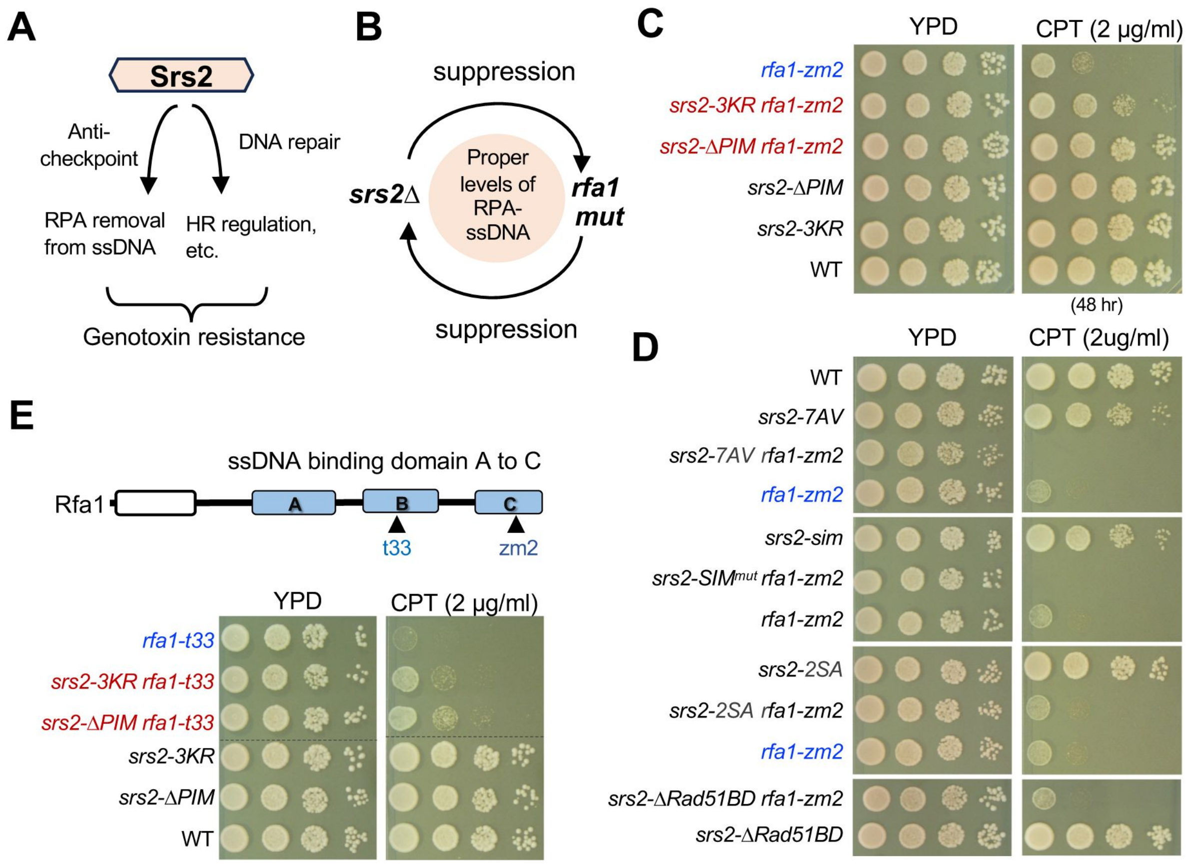

Figure 2.

Srs2 PCNA binding and sumoylation are involved in suppressing *rfa1* mutants

(A) Schematic to show that Srs2's roles in checkpoint dampening (anti-checkpoint for simplicity) and DNA repair both contribute to genotoxin resistance.

(B) Schematic to highlight the mutual suppression between *srs2Δ* and *rfa1* mutants and the underlying effects on the RPA-ssDNA levels in cells.

(C) *srs2-ΔPIM* and *-3KR* suppress *rfa1-zm2* sensitivity to CPT. A 10-fold serial dilution of cells of the indicated genotypes were spotted and growth was assessed after incubation at 30°C for 40 hours unless otherwise noted.

(D) Several *srs2* mutants did not suppress *rfa1-zm2* sensitivity to CPT. Experiments were performed as in panel 2C.

(E) *srs2-ΔPIM* and *-3KR* suppress *rfa1-t33* sensitivity to CPT. Top: schematic of the Rfa1 protein domains and *rfa1* mutant alleles. Bottom: *srs2-ΔPIM* and *-3KR* improved *CPT* sensitivity of *rfa1-t33* cells. Experiments were performed as in panel 2C.

suppression, consistent with the notion that Srs2 has roles beyond RPA regulation (Bronstein et al., 2018; Niu & Klein, 2017). Nevertheless, the observed rescue indicates that Srs2 PCNA binding and sumoylation are likely involved the regulation of RPA during the DNA damage response.

Srs2 regulation of RPA during DDC downregulation is not only observed under CPT treatment, but also in MMS treated conditions (Dhingra et al., 2021). Consistent with this observation, *srs2-ΔPIM* moderately improved MMS sensitivity of *rfa1-zm2* and *rfa1-t33*, while *srs2-3KR* had a small effect on the MMS sensitivity of *rfa1-zm2* (Fig. S1B, S1C). Collectively, the genetic results provide evidence that the roles of Srs2 binding to PCNA and sumoylation in RPA antagonism can be applicable to more than one genomic stress situations.

srs2-ΔPIM and *-3KR* rescue checkpoint abnormality of *rfa1-zm2* cells

We next investigated whether *srs2-ΔPIM* and *-3KR* suppression of *rfa1-zm2* pertained to DNA damage checkpoint regulation. We employed two readouts for DDC, including the levels of the phosphorylated and active form of Rad53 detectable by the F9 antibody (Fiorani et al., 2008) and cellular ability to exit the G2/M phase after CPT treatment. We have previously shown that *srs2Δ* cells exhibit increased levels of Rad53 phosphorylation and diminished ability to exit G2/M phase in CPT, and that both are suppressed by *rfa1-zm2* (Dhingra et al., 2021). Interestingly, *rfa1-zm2* leads to similar defects and these are rescued by *srs2Δ* (Dhingra et al., 2021). The mutual suppression seen in these assays is consistent with that seen for CPT sensitivity (Dhingra et al., 2021). We note that several other RPA ssDNA binding mutants, such as *rfa1-t33*, also exhibit phenotypes indicative of increased DDC levels (Pellicioli et al., 2001; Seeber et al., 2016; Smith & Rothstein, 1995; Lee Zou & Stephen J. Elledge, 2003). This is likely because compromised ssDNA protection can lead to increased levels of DNA lesions thus enhancing the activation of Mec1 and/or its homolog Tel1.

We reasoned that if *srs2-ΔPIM* and *srs2-3KR* compromise Srs2's ability to remove RPA from DNA in cells, they should alleviate *rfa1-zm2* cells' defects in ssDNA protection and the associated DDC abnormalities. Indeed, while *rfa1-zm2* exhibited about 5-fold increase in Rad53 phosphorylation levels in CPT compared with wild-type cells as seen before (Dhingra et al., 2021), *srs2-ΔPIM* and *-3KR* reduced this level by about 50% and 80%, respectively (**Fig. 3A**). Similar to earlier data, we observed that wild-type cells arrested in G_2/M after 1 hour of CPT treatment and about 25% of these cells entered G_1 after another hour (**Fig. 3B**) (Dhingra et al., 2021). The ability to exit from G_2/M phase was reduced in *rfa1-zm2* cells so that only 13% cells moved to G1 phase after 2 h treatment with CPT, as seen before (**Fig. 3B**) (Dhingra et al., 2021). Significantly, *srs2-ΔPIM* and *-3KR* increased the percentage of *rfa1-zm2* cells transitioning into the G1 phase (**Fig. 3B**). The suppressive effects seen in both assays were stronger for *srs2-ΔPIM* than *srs2-3KR* as seen in genotoxicity assays (Fig. S1B; 3A, 3B). Collectively, the genetic and DDC assay data are consistent with each other and suggest that Srs2 binding to PCNA and its sumoylation contribute to RPA antagonism in DDC regulation.

srs2-ΔPIM is additive with a *slx4* mutant defective in DDC recovery

We found that *srs2-ΔPIM* and *srs2-3KR* mutants on their own behaved normally in the two DDC assays described above, which differs from *srs2Δ* cells that exhibit increased levels of Rad53 phosphorylation and reduced G1 entry (**Fig. 3A**, 3B) (Dhingra et al., 2021). One plausible interpretation of this difference is that unlike *srs2*Δ, these partial *srs2* mutants only mildly impair Srs2-mediated RPA regulation during DDC downregulation, such that their defects can be compensated for by other DDC dampening systems. To test this idea, we examined the consequences of removing the Slx4-mediated anti-checkpoint function in *srs2-ΔPIM* cells. Slx4 is known to bind the scaffold protein Rtt107, and their complex can compete with the DDC protein Rad9 for binding to damaged chromatin, consequently disfavoring Rad9's role in promoting Rad53 phosphorylation (Ohouo et al., 2010; Ohouo et al., 2013). Slx4 uses an Rtt107 interaction motif

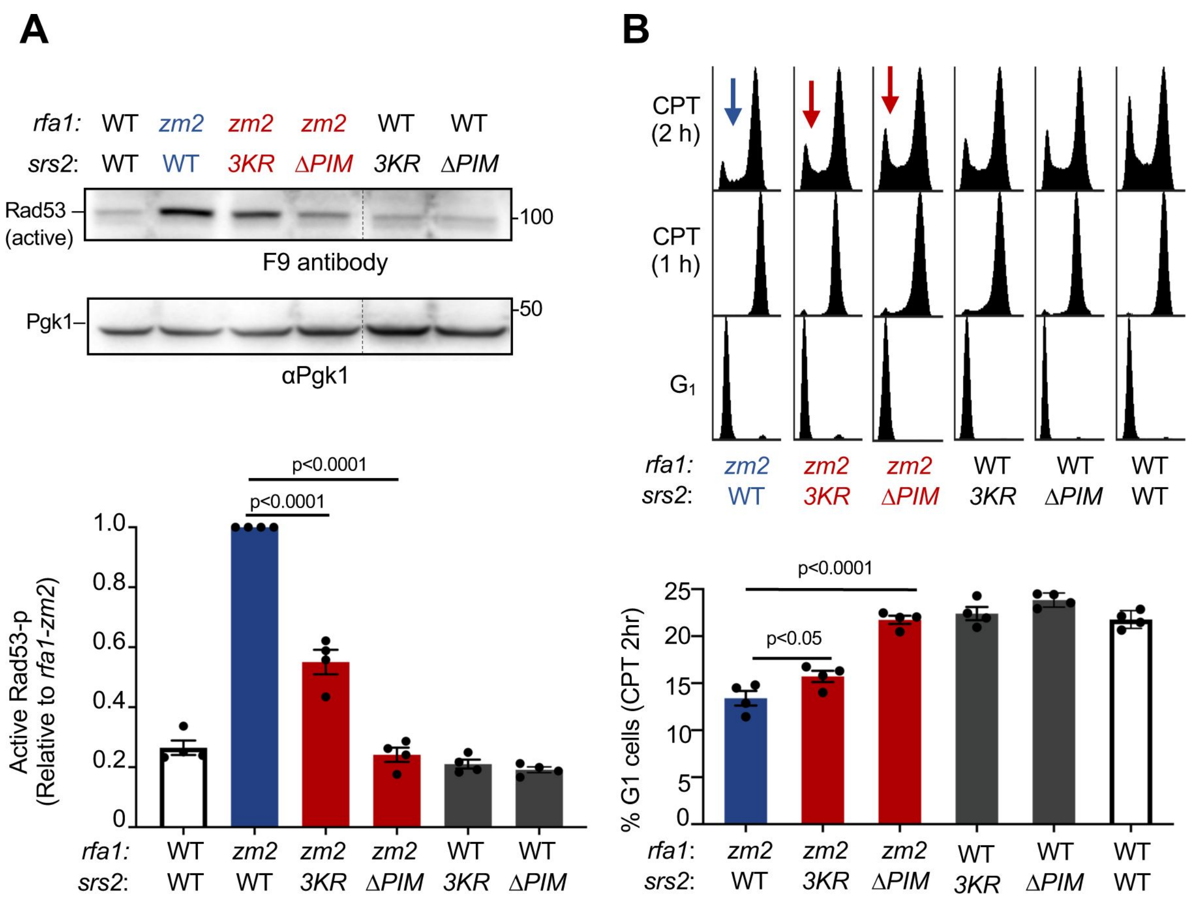

Figure 3.

***srs2-ΔPIM* and *-3KR* correct checkpoint abnormalities in *rfa1-zm2* cells.**

(A) *srs2-ΔPIM* and *-3KR* reduce the levels of active Rad53 in *rfa1-zm2* cells. Protein extracts were examined after G1 cells were released into cycling in the presence of 16 ug/ml CPT for 2 hours. Activated Rad53 was detected by the F9 antibody by immunoblotting. Active Rad53 signals in each indicated strain were compared to the Pgk1 loading control and normalized to those of *rfa1-zm2*. Two biological isolates per genotype were examined in two technical repeats and results are graphed with error bars representing SD. T-test generated *p* value is indicated.
(B) *srs2-ΔPIM* and *-3KR* allow better G1 entry of *rfa1-zm2* cells. Experiments were performed as in panel A and samples were collected at indicated time points. FACS profiles of the samples are shown at the top and percentages of G1 cells after 2 hours of CPT treatment (CPT 2 h) are plotted at the bottom. Two biological isolates per genotype were examined in two technique repeats and results are graphed with error bars representing SD. T-test generated *p* value is indicated.

(RIM) to bind Rtt107 and a previously established RIM mutant, *slx4*^{RIM} (T423A, T424A, and S567A), abolishes Rtt107 binding without affecting other Slx4 protein attributes (Wan et al., 2019). *slx4*^{RIM}was tagged with a TAP tag at its C-terminal end and examined together with a control wherein the wild-type Slx4 was tagged in the same manner. Compared with the control, *slx4*^{RIM}exhibited mild sensitivity toward CPT (Fig. S2A). Significantly, when *srs2-ΔPIM* was combined with *slx4*^{RIM}, the double mutant exhibited stronger CPT sensitivity compared with corresponding single mutants (Fig. S2A).

We further examined DDC levels when cells were exposed to CPT using both Rad53 phosphorylation and ability to exit G2/M as readouts. Compared with *Slx4-TAP* cells, *slx4*^{RIM}*-TAP* cells exhibited 2-fold increase in Rad53 phosphorylation level, consistent with a role of RIM of Slx4 in downregulating DDC (Fig. S2B). In *Slx4-TAP* background, *srs2-ΔPIM* led to 3-fold increase in Rad53 phosphorylation level (Fig. S2B). As cells containing untagged Slx4 and *srs2-ΔPIM* did not show Rad53 phosphorylation changes (**Fig. 3A**), tagging Slx4 likely sensitizes *srs2-ΔPIM.* Significantly, the double mutant of *srs2-ΔPIM* and *slx4*^{RIM}*-TAP* exhibited further increase in Rad53 phosphorylation (~5.5 fold) compared with *srs2-ΔPIM* and *slx4*^{RIM}*-TAP* single mutant (Fig. S2B). Congruous with this finding, the *srs2-ΔPIM slx4*^{RIM}*-TAP* double mutant also reduced the percentage of cells entering G_1 from G2/M arrest compared with their single mutants (Fig. S2C). Together, these three lines of evidence support the notion that defects in the Srs2-mediated DDC downregulation can be compensated for by Slx4-mediated mechanisms.

Srs2 binding to PCNA facilitates Srs2 sumoylation

We next addressed whether the two Srs2 features involved in RPA antagonism are functionally related. To this end, we tested if Srs2 binding to PCNA affected Srs2 sumoylation in cells. We confirmed that endogenous sumoylation of Srs2 could be detected in cell extracts after enriching SUMOylated proteins using 8His-tagged SUMO (8His-Smt3) (**Fig. 4A**). Briefly, proteins were extracted from cells under denaturing conditions to disrupt protein-protein association and prevent desumoylation, and sumoylated proteins were enriched on Ni-NTA (nickel-nitrilotriacetic acid) resin (Ulrich & Davies, 2009). Consistent with previous findings, sumoylated forms of Srs2 in the Ni-NTA eluate were detected on immunoblots using an anti-Srs2 antibody only when cells contained 8His-Smt3, where unmodified Srs2 was seen regardless of the presence or absence of 8His-Smt3 due to nonspecific binding to the resin (**Fig. 4A**) (Saponaro et al., 2010). These bands were only seen in wild-type cells but not *srs2Δ* cells (Fig. S3A). We found that Srs2 sumoylation level was reduced in *srs2-3KR* strains in CPT conditions, as expected from mutating the three major sumoylation sites (**Fig. 4A**) (Saponaro et al., 2010). Interestingly, *srs2-ΔPIM* cells exhibited a similar reduction in Srs2 sumoylation as seen in *srs2-3KR* cells (**Fig. 4A**). This result suggests that Srs2 sumoylation largely depends on Srs2 binding to PCNA, supporting the notion that the two events occur sequentially.

Srs2 sumoylation peaks after maximal Mec1 activation and depends on Mec1

While RPA can engage with ssDNA generated in different processes, such as transcription and replication, only a subset of ssDNA regions is adjacent to 3'-junctions loaded PCNA. This is because PCNA loading occurs mainly during S phase and requires an ssDNA and dsDNA junction site permissive for loading (Moldovan, Pfander, & Jentsch, 2007). We thus reasoned that Srs2 binding to PCNA may prevent Srs2 from removing RPA bound to ssDNA that forms transiently or is not flanked by PCNA, such as within transcription bubbles, R-loop regions, or at ongoing replication forks. This could provide a means for spatial control of Srs2's anti-RPA role. We explored whether this role could also be regulated temporally. As the genetic data described above suggest that Srs2 sumoylation acts downstream of Srs2 binding to PCNA in antagonizing RPA, we examined the timing of Srs2 sumoylation after CPT treatment.

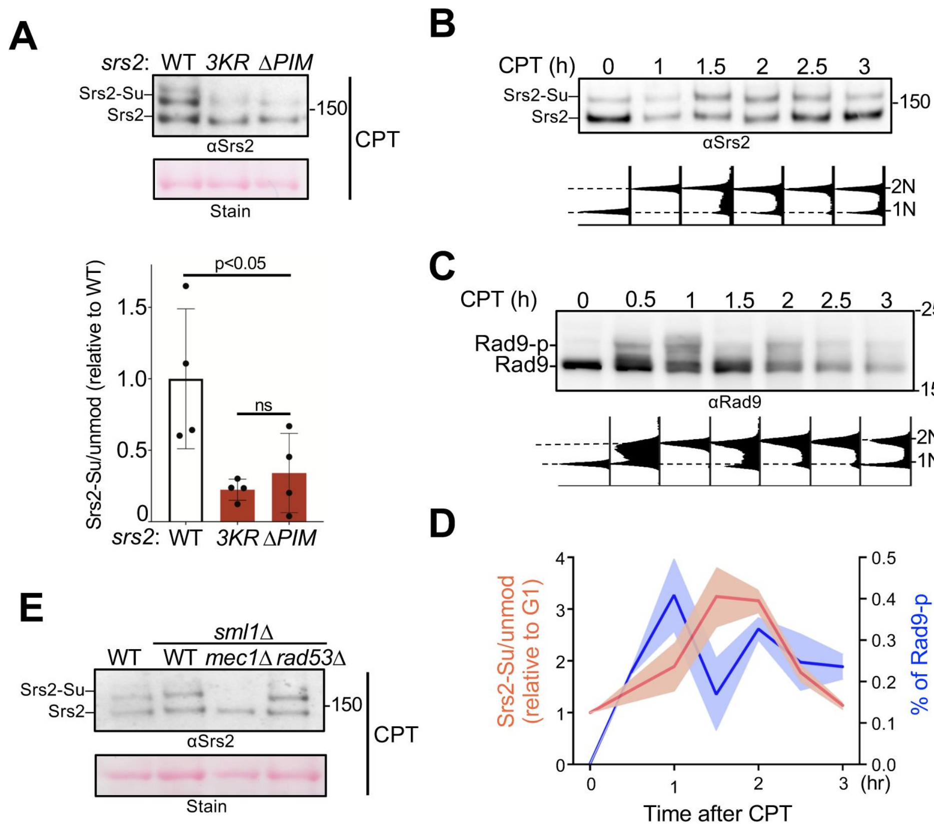

Figure 4.

Srs2 sumoylation depends on Srs2 PIM motif and the Mec1 kinase

(A) Srs2 sumoylation level is reduced in *srs2-ΔPIM* cells. Samples were collected after CPT treatment as in **Figure 3A** and sumoylated proteins were enriched as described in the text and methods. Sumoylated form of Srs2 (Srs2-Su) migrated slower than unmodified Srs2 on gel and both were detected using anti-Srs2 antibody in immunoblotting. Equal loading is indicated by Ponceau-S stain (Stain). The levels of sumoylated Srs2 relative to those of unmodified Srs2 were normalized to wild-type cells and plotted. Two biological isolates per genotype were examined in two technique repeats and results are graphed with error bars representing SD. T-test generated *p* value is indicated. ns, not significant.

(B) Srs2 sumoylation level during CPT treatment. Samples were collected as described in **Figure 3B**. Experiments were conducted, and immunoblotting data is presented as those in panel A. Time 0 samples are G1 cells before CPT treatment. FACS analysis at corresponding time points are included below the plot.

(C) Rad9 phosphorylation level during CPT treatment. Samples used in panel B were examined for Rad9 phosphorylation using an anti-Rad9 antibody. Unmodified and phosphorylated (Rad9-p) Rad9 forms are indicated. FACS analysis at corresponding time points are indicated below blots.

(D) Quantification of Srs2 sumoylation and Rad9 phosphorylation level changes during CPT treatment. Srs2 sumoylation signals were normalized based on unmodified form. Values at each timepoint were relative to that of G1 sample (Time 0) before CPT treatment, and fold changes were indicated on left y-axis. Rad9 signals were normalized to non-modified form and percentage of phosphorylated form were indicated on right y-axis. Two biological isolates per genotype were examined in two technique repeats and results are graphed with error bars representing SD.

(E) Mec1 is required for Srs2 sumoylation. Experiments were performed and data are presented as in panel A.

We released G1 synchronized cells into media containing CPT and examined Srs2 sumoylation at several time points (**Fig. 4B**). A similar experimental scheme was used to monitor the level of Mec1 activation using Rad9 phosphorylation as a readout (**Fig. 4C**). Quantification of Srs2 sumoylated forms relative to its unmodified forms showed that its sumoylation level peaked between 1.5 and 2 h after cells were released into CPT (**Fig. 4D**). Quantification of phosphorylated forms of Rad9 relative to its unmodified form showed that checkpoint activation peaked 1 hour after cell release (**Fig. 4D**). That checkpoint level peaked before Srs2 sumoylation level peaked suggests a possible dependence of the latter on the checkpoint. In testing this idea, we found that *mec1Δ* cells, which contain *sml1Δ* to support viability, abolished Srs2 sumoylation (**Fig. 4E**). Neither *sml1Δ* alone nor removal of Rad53 affected Srs2 sumoylation (**Fig. 4E**), suggesting that Mec1's effect on this modification does not require the downstream Rad53 effector kinase. As Mec1 is not generally required for protein sumoylation during the DNA damage response (Cremona et al., 2012), its involvement in Srs2 sumoylation presents a rather unique effect.

Examination of Mec1-S1964 phosphorylation in Srs2-mediated DDC regulation

The combined results that Srs2 sumoylation levels peak after those of Rad9 phosphorylation and that Srs2 sumoylation depends on Mec1 raised the possibility that Mec1 hyper-activation during the late part of the DNA damage response may favor Srs2's role in DDC dampening. Recently, Mec1 protein bound to Ddc2 was shown to be auto-phosphorylated at Ser1964 in a late part of the DNA damage response after a single DNA break is induced by the HO endonuclease (Memisoglu et al., 2019). Further, the phosphorylation-defective mutant *mec1-S1964A* impaired checkpoint downregulation in this condition, leading to the model that Mec1-S1964 phosphorylation contributes to DDC dampening at least after HO-induced break (Memisoglu et al., 2019). We tested the idea that Mec1-S1964 phosphorylation is also involved in Srs2-mediated DDC regulation under CPT conditions.

First, we asked whether Mec1-S1964 phosphorylation occurs under CPT conditions using a time course scheme similar to the one used in testing Srs2 sumoylation. Ddc2-bound Mec1 was recovered after immunoprecipitating Ddc2 at each timepoint and the proteins were examined by immunoblotting. To detect Mec1-S1964 phosphorylation, we used an antibody raised against the peptide containing this modification (Memisoglu et al., 2019). The antibody detected the Mec1 S1964 phosphorylation band in immunoprecipitated fractions only when Ddc2 was tagged with the Myc tag but not when Ddc2 was untagged (Fig. S3B). Further, the Mec1 S1964 phosphorylation band was largely abolished in the *mec1-S1964A* cells. (Fig. S3B; 5A). Given that the antibody also detects a closely spaced background band (Fig. S3B; 5A), we calculated the relative level of Mec1 S1964 phosphorylation in each sample by normalizing to this band. Quantification results showed that Mec1 S1964 phosphorylation level peaked around the same time as seen for Rad9 phosphorylation, which is 1 hour after CPT treatment (**Fig. 5A**).

After confirming that Mec1 S1964 phosphorylation also took place under CPT conditions, we tested whether the *mec1-S1964A* or the phosphorylation-mimetic mutant, *mec1-S1964E*, influenced *rfa1-zm2* or *srs2* growth on media containing CPT. If Mec1 S1964 phosphorylation aids Srs2-mediated RPA antagonism in CPT, we would expect *mec1-S1964A* to behave similarly to *srs2* mutants in rescuing *rfa1-zm2* CPT sensitivity, while *mec1-S1964E* should have the opposite effects. We found that *mec1-S1964A*, which showed normal CPT resistance on its own, did not affect the growth of *rfa1-zm2* cells on CPT-containing meidua (**Fig. 5B**). *mec1-S1964A* also did not influence the growth of *srs2-ΔPIM* and *srs2-3KR* cells regardless of *rfa1-zm2* (**Fig. 5B**, S5A).

Though *mec1-S1964E* on its own did not cause CPT sensitivity, it led to slight but reproducible slow growth in cells containing *srs2-3KR* and *srs2-ΔPIM* with or without *rfa1-zm2* (Fig. S4B). We found that neither *mec1-S1964A* nor *mec1-S1964E* affected Srs2 sumoylation levels (Fig. S4C). Thus, Mec1, but not its S1964 phosphorylation, is required for Srs2 sumoylation. These data provide evidence

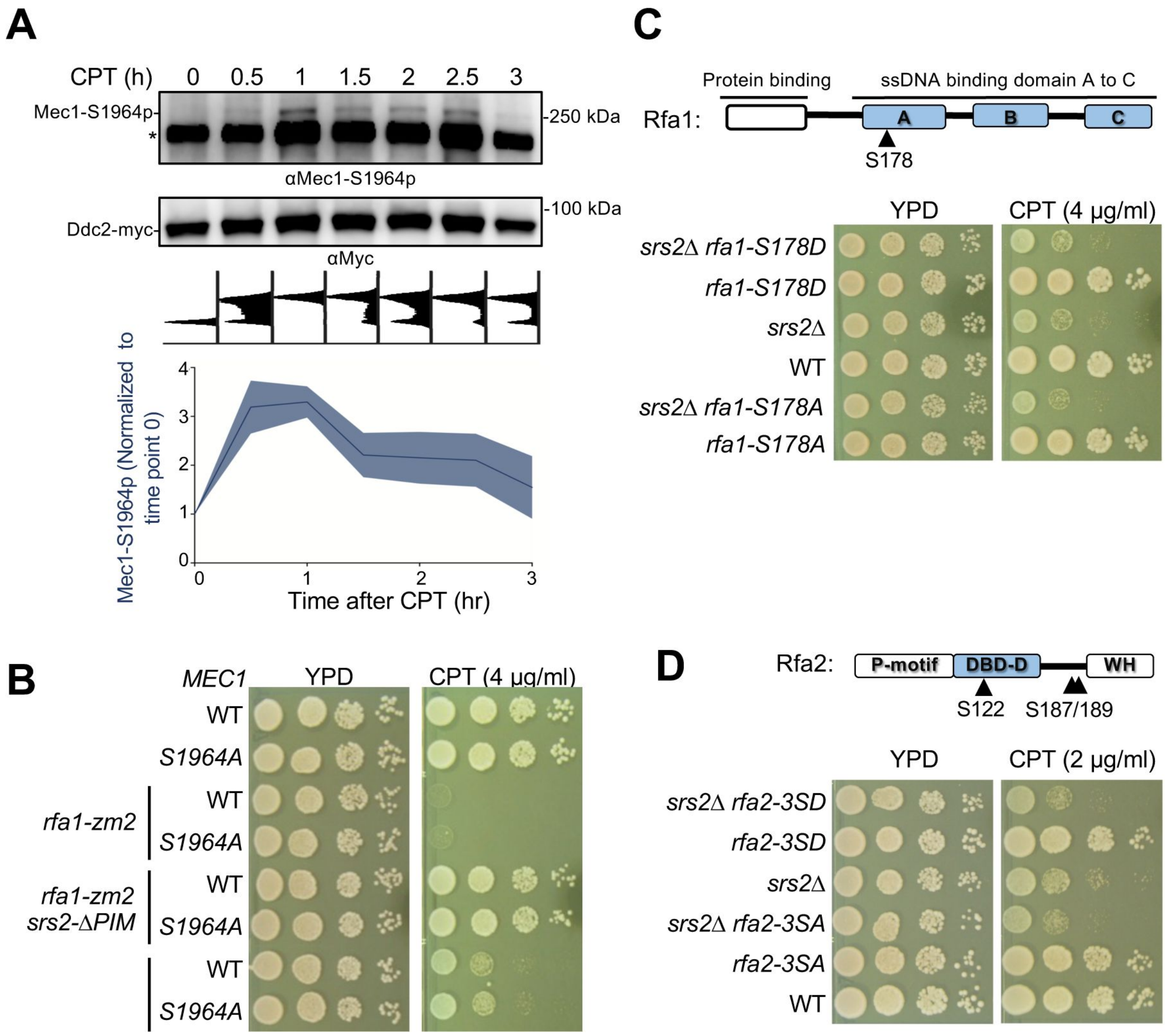

Figure 5.

Mec1 and RPA phosphorylation did not link to Srs2 anti-checkpoint function.

(A) Mec1 autophosphorylation at S1964 during CPT treatment. Samples were collected as described in **Figure 3B**. Ddc2-myc was immunoprecipitated and the associated Mec1 was co-purified. The anti-Mec1-S1964-p antibody detected the phosphorylated form of Mec1 and a non-specific band (*). Quantification below the blot was based on results from two biological isolates.

(B) *mec1-S1964A* did not rescue *rfa1-zm2* sensitivity toward CPT. Experiments were performed as in **Figure 2C**.

(C) Examination of the *rfa1-S178A* and *-S178D* mutants. Top: schematic of the Rfa1 protein domains and position of Ser178. ssDNA binding domain (DBD) A-C are shown. Bottom, *rfa1-S178A* and *-S178D* had no effect on *srs2Δ*'s sensitivity to CPT. Experiments were performed as in **Figure 2C**.

(D) Examination of the *rfa2-3SA* and *-3SD* mutants. Top, schematic of the Rfa2 protein domains and position of Ser122, Ser187 and Ser189 mutated in *rfa2-3SA* and *-3SD*. ssDNA binding domain (DBD)-D is indicated. Bottom, *rfa2-3SA* has no effect on *srs2Δ*'s sensitivity toward CPT while *rfa2-3SD* showed additive effect. Experiments were performed as in **Figure 2C**.

that even though an HO-induced DNA break and CPT treatment both induce Mec1 S1964 phosphorylation, this modification appears to not affect Srs2-mediated DDC dampening in the former situation.

Genetic data suggest that known RPA phosphorylation sites do not affect DDC recovery

While we mainly focused on the identification of Srs2 features that contribute to RPA antagonism during DDC regulation, we also considered the possibility that certain regulatory RPA features may contribute to this regulation as well. In particular, RPA is known to be phosphorylated by Mec1 (Chen et al., 2010; Faca et al., 2020; Lanz et al., 2021). We thus tested whether this phosphorylation may render RPA more susceptible to be removed from ssDNA by Srs2. To this end, we mutated previously identified Mec1 phosphorylation sites on the Rfa1 and Rfa2 subunits. These include Rfa1 Ser178 and three sites on Rfa2 (S122, S187, S189) (Albuquerque et al., 2015; Chen et al., 2010). We generated corresponding non-phosphorylatable mutants, *rfa1-S178A* and *rfa2-3SA (*S122A, S187A, and S189A), and phosphomimetic mutants, *rfa1-S178D* and *rfa2-3SD*.

Using genetic suppression as a readout for possible involvement of RPA mutants in Srs2-based DDC regulation, we examined the interactions between these mutants and *srs2*Δ. If RPA phosphorylation promote its removal by Srs2, the *SD* mutants may show suppressive interactions with *srs2*Δ, while *SA* mutants would behave like *srs2* mutants. We found that neither *rfa1-S178D* nor *S178A*, both of which showed normal CPT resistance on their own, affected *srs2*Δ cell growth on CPT-containing media (**Fig. 5C**). In contrast, *rfa2-3SD* or *rfa2-3SA,* which also behaved like wild-type regarding CPT sensitivity, sensitized *srs2Δ* growth on CPT-containing media (**Fig. 5D**). This negative genetic interaction is in contrast to the positive interaction seen between *srs2*Δ and *rfa1-zm2* or *rfa1-t33* (Fig. S1A) (Dhingra et al., 2021). The differential Srs2 genetic interactions among distinct RPA alleles likely reflect the multifunctional natures of Srs2 and RPA. Collectively, these analyses suggest that the known phosphorylation sites on RPA are unlikely to be involved in Srs2-based DDC regulation.

Discussion

Timely termination of DDC is crucial for cell survival, yet the underlying mechanisms are not well understood. Using the budding yeast model system, we investigated how the Srs2-mediated DDC dampening through antagonizing RPA can be regulated. Our previous work reports a requirement of Srs2's ATPase activity in this role; here we examined six additional Srs2 features involved in either binding to other proteins or post-translational modifications. Our results suggest that among the examined features, only Srs2 binding to PCNA and its sumoylation are involved in RPA antagonism during DDC downregulation. Interestingly, these two features are functionally linked, as optimal Srs2 sumoylation requires Srs2 binding to PCNA. Further, the Srs2 sumoylation level peaks after Mec1 activity reaches its maximum and that Mec1 is required for this modification. We posit a model that can interpret both previous and current data (**Fig. 6**). We envision that Srs2 binding to PCNA that is loaded at the 3' junction site flanking ssDNA gaps help to bring Srs2 in proximity to the RPA-ssDNA filament and promote Srs2 sumoylation, both favoring downregulation of the checkpoint (**Fig. 6**). These regulatory mechanisms could in principle help to minimize undesirable RPA loss from PCNA-free regions and premature RPA removal from DNA during checkpoint induction phase.

Our initial assessment of six Srs2 mutants used genetic suppression of the *rfa1-zm2* CPT sensitivity as a readout for Srs2 and RPA antagonism. The results suggest that RPA regulation involves Srs2 binding to PCNA and Srs2 sumoylation, but not Srs2 binding to either Rad51 or SUMO nor its phosphorylation at Cdk1 sites and two potential Mec1 kinase sites. Genetic analysis showed that the suppression conferred by mutating PCNA binding sites on Srs2 (*srs2Δ-PIM*) in *rfa1-zm2* cells

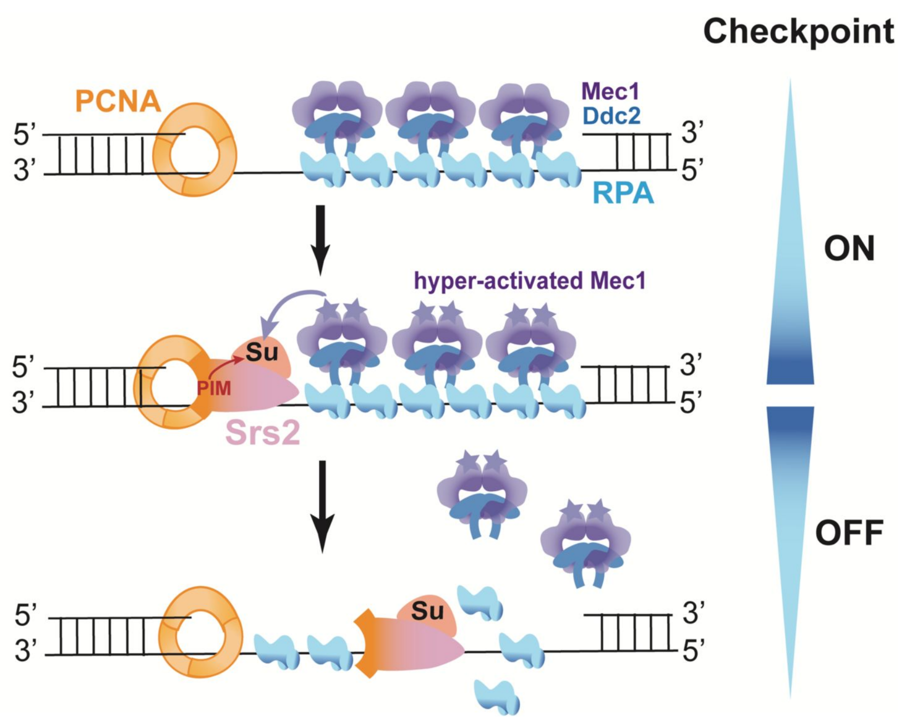

Figure 6.

A working model for the regulation of Srs2-mediated checkpoint dampening.

Genomic sites containing ssDNA gaps can be generated during disrupted DNA replication and DNA repair under genotoxin treatment. The resultant ssDNA gaps can be coated by RPA, which then recruit the Mec1/Ddc2 kinase complex. Mec1 is subsequently activated by DDC proteins, such as the 9-1-1 complex that demarcates the 5’ end junctions of ssDNA gaps (not drawn).

After DDC is activated for a prolonged period, Srs2 recruited to PCNA that demarcates the 3’ end junction flanking the ssDNA gaps can be sumoylated. Srs2 sumoylation level increases as Mec1 activation heightens and can result in more efficient RPA removal, leading to Mec1 downregulation.

also extended to MMS conditions. Additionally, a similar suppression for another *rfa1* mutant defective in ssDNA binding was seen. These results ruled out suppression being specific to one allele or one genotoxin. Data from two checkpoint assays further showed that *srs2Δ-PIM* and *srs2-3KR* suppressed DDC abnormalities in *rfa1-zm2* cells, strengthening the conclusion that Srs2 binding to PCNA and its sumoylation help to counter RPA during the DNA damage response.

RPA coats ssDNA generated from many DNA metabolic processes; thus, selective RPA stripping can be beneficial in cells. Our data suggest that this selectivity could be partially achieved by Srs2 binding to PCNA. In principle, ssDNA regions not adjacent to stable 3' ss-dsDNA junction sites, such as those formed during transcription and in the R-loop regions, could be spared from Srs2-mediated RPA stripping. In contrast, ssDNA regions adjacent to PCNA-loaded junction sites, such as at ssDNA gap regions behind the replication forks, can be better targeted for Srs2-mediated RPA removal. An involvement of PCNA binding in Srs2-based RPA regulation could also compensate for a lack of Srs2 interaction with RPA. This scenario differs from Srs2-based Rad51 stripping that requires their interaction (Antony et al., 2009; Colavito et al., 2009; Krejci et al., 2003). It is thus likely that though both the anti-recombinase and anti-checkpoint roles of Srs2 can benefit from selectivity, they are achieved via different means. Previous studies have suggested that both PIM and SIM aid Srs2 recruitment to sumoylated form of PCNA (Papouli et al., 2005; Pfander et al., 2005). Our data reveal that at least in CPT conditions, Srs2-PIM, but not Srs2-SIM, is involved in RPA regulation. This observation is congruent with phenotypic differences between Srs2's PIM and SIM seen in other studies and here, such as much stronger genotoxic sensitivity of the latter (Fig. S1A) (Fan et al., 2023; Kolesar et al., 2016; Kolesar et al., 2012). Considering that only a small percentage of PCNA is sumoylated and mainly during S phase (Papouli et al., 2005; Pfander et al., 2005), it is probably advantageous for Srs2 not being restricted to sumoylated PCNA for efficient RPA regulation.

How could Srs2 binding to PCNA helps Srs2 sumoylation? While a full answer awaits future studies, we speculate that this interaction may render Srs2 more permissive for sumoylation and/or position Srs2 in proximity with its sumoylation E3 ligase Siz2 that binds RPA (Chung & Zhao, 2015; Kolesar et al., 2012). Considering that Mec1, Ddc2, and RPA contain SUMO interaction motifs (Psakhye & Jentsch, 2012), it is possible that Srs2 sumoylation could help enrich the helicase at ssDNA-RPA sites via interacting with these motifs, though other effects are also possible. Compared with *srs2-3KR*, *srs2-ΔPIM* showed better suppression of *rfa1-zm2* defects. This indicates that during RPA regulation, PCNA plays additional roles besides promoting Srs2 sumoylation.

We found that Srs2 sumoylation in cells also requires Mec1, which does not promote sumoylation in general (Cremona et al., 2012). On the contrary, Mec1 loss causes increased sumoylation levels (Cremona et al., 2012). It is currently unclear how Mec1 can specifically promote Srs2 sumoylation, but this may involve Mec1-mediated phosphorylation of SUMO substrates and/or enzymes. A requirement of Mec1 for Srs2 sumoylation and the dynamic nature of sumoylation suggest that Srs2 sumoylation may be a potential timing regulator. It is possible that hyperactivation of Mec1 favors Srs2 sumoylation that may aid its own downregulation. While fully testing this idea requires time-course phosphoproteomic studies under genotoxin treatment, we examined a Mec1 auto-phosphorylation site implicated in its downregulation after induction of a single DNA break (Memisoglu et al., 2019). We did not find evidence that this is involved in Mec1 dampening in CPT condition, raising the possibility that distinct mechanisms may be used in different genomic stress conditions. We also found that two phosphorylation sites on Srs2 that showed Mec1 dependency in proteomic studies had no effect in the CPT situation (Albuquerque et al., 2015; Faca et al., 2020); however we note that these sites do not fit with the typical S/TQ Mec1 consensus sites, thus may not be the bona fide Mec1 sites. Other phosphorylation sites examined here include five Mec1 sites on Rfa1 and Rfa2 (Albuquerque et al., 2015; Chen et al., 2010). Genetic analyses did not support their roles in the Srs2 and RPA antagonism. Thus, future studies to comprehensively map DDC kinase phosphorylation sites on Srs2 and RPA, as well as their regulators, such as the RPA chaperone Rtt105 (Li et al., 2018), will be needed to test how the kinases can communicate with Srs2-based RPA regulation during the DNA damage response.

While we provide multiple lines of evidence in support of the involvement of Srs2 binding to PCNA and its sumoylation in RPA regulation during checkpoint downregulation, *srs2-ΔPIM* or *-3KR* alone behaved normally in checkpoint assays. The lack of a defect here can be due to buffering effects, both from not-yet characterized Srs2 features and from other checkpoint dampening pathways. We provided evidence for the latter since *srs2-ΔPIM* or *-3KR* showed additive phenotype when combined with a *slx4* mutant defective in DDC dampening function. The presence of multiple checkpoint dampening factors that target different checkpoint modules highlights the importance of the process. Recent demonstration of other DNA helicases, such as the human HELB, HELQ, and BLM proteins, in stripping of RPA from ssDNA in vitro suggests that Srs2-like checkpoint downregulation may present in human cells as well (Hormeno et al., 2022; Jenkins et al., 2021; Shorrocks et al., 2021). Since RPA-ssDNA binding not only plays a key role in checkpoint, but also in many other processes, the Srs2 features uncovered here may help to better understand how RPA dynamics is controlled by helicases and other factors during additional processes beyond DDC.

Methods

Yeast strains and genetic techniques

Standard procedures were used for cell growth and media preparation. The strains used in this work are listed in Table S1and are isogenic to W1588-4C, a *RAD5* derivative of W303 (Zhao & Blobel, 2005). Mutant alleles of *srs2*, *rfa1*, *rfa2* and *mec1* used in this work were generated by CRISPR-Cas9-based gene replacement, except for *srs2-3KR* that was introduced using a PCR-based method (DiCarlo et al., 2013). All mutations were verified by sequencing. Standard procedures were used for tetrad analyses and genotoxin testing. For each assay, at least two biological duplicates were used per each genotype. Cells were grown at 30 °C unless otherwise stated.

Cell synchronization and cell cycle analyses

Cells from log-phase cultures were treated with 5 μg/mL α factor (GenScript RP01002) for 1 h, followed by an additional 2.5 μg/mL α factor for 30 min. When at least 95% cells were arrested in G1 phase based on the percentage of unbudded cells, they were released into yeast extract–peptone–dextrose (YPD) media containing 100 μg/mL Protease (Millipore 53702) and 16 μg/mL CPT (Sigma C9911) for 2 h. Cell cycle progression was monitored by standard flow cytometry analyses as described previously (Zhao & Rothstein, 2002).

Protein extraction using trichloroacetic acid (TCA)

To examine the protein levels of Srs2, the phosphorylation form of Rad53, and Rad9, cell extracts were prepared as reported (Regan-Mochrie et al., 2022) . Briefly, 2×10^8 cells were collected at indicated time points. Cell pellets were resuspended in 20% TCA and lysed by glass bead beating. The lysate was then centrifuged to remove supernatant. Precipitated proteins were suspended in Laemmli buffer (65 mM Tris-HCl at pH 6.8, 2% SDS, 5% 2-mercaptoethanol, 10% glycerol, 0.025% bromophenol blue) with 2 M Tris added to neutralize the solution. Prior to loading, samples were boiled for 5 min and spun down at 14,000 rpm for 10 min to remove insoluble materials. Samples were separated on 4-20% Tris-glycine gels (Bio-Rad 456-1096) to examine Srs2 level and active Rad53 form or 7% Tris-acetate gels (Thermo Fisher EA03555) to detect Rad9 phosphorylation.

Detection of Srs2 sumoylation

Sumoylated proteins were pulled down from cells containing 8His-tagged SUMO expressed from its endogenous locus using the standard Ni-NTA pull-down method as previously described (Ulrich & Davies, 2009). Briefly, protein extracts prepared in 55% TCA were incubated in buffer A (6 M guanidine HCl, 100 mM sodium phosphate at pH 8.0, 10 mM Tris-HCl at pH 8.0) with rotation for 1 h at room temperature. The cleared supernatant was obtained after centrifuging for 20 min and was then incubated overnight at room temperature with Ni-NTA resin (Qiagen 30210) in the presence of 0.05% Tween-20 and 4.4 mM imidazole with rotation. Beads were washed twice with buffer A supplemented with 0.05% Tween 20 and then four times with buffer C (8 M urea, 100 nM sodium phosphate at pH 6.3, 10 mM Tris-HCl at pH 6.3) supplemented with 0.05% Tween 20. Proteins were eluted from the beads using HU buffer (8 M urea, 200 mM Tris-HCl at pH 6.8, 1 mM EDTA, 5% SDS, 0.1% bromophenol blue, 1.5% DTT, 200 mM imidazole). Samples were loaded onto NuPAGE^{TM} 3– 8% Tris-acetate gels (Thermo Fisher EA03752) for immunoblotting to detect both sumoylated and unmodified Srs2. Equal loading was verified using Ponceau S staining.

Detection of Mec1 S1964 phosphorylation during CPT treatment

G1-arrested cells were filtered and resuspended in prewarmed YPD media containing 16 μg/mL CPT to allow entry into S phase. Samples were collected at indicated time points. Cells were disrupted by glass bead beating in lysis buffer, followed by centrifugation at 20,000 g for 15 min to obtain whole-cell extract. The lysis buffer consisted of 25 mM K-HEPES (pH 7.6), 100 mM NaCl, 100 mM K-glutamate, 5 mM $Mg(OAc)_2$, 0.02% NP40, and 0.5% Triton X-100 supplemented by protease inhibitor cocktail (Sigma P8215) and Complete Ultra EDTA-free protease inhibitor (Roche 11836145001). The activated form of Mec1 was enriched by immunoprecipitating myc-tagged Ddc2 using Protein A beads and an anti-Myc antibody (Thermo Fisher 9E10) for 2–4 h at 4°C. Beads were washed five to six times with lysis buffer, and proteins were eluted using Laemmli buffer (65 mM Tris-HCl at pH 6.8, 2% SDS, 5% 2-mercaptoethanol, 10% glycerol, 0.025% bromophenol blue). After boiling for 5 min, eluted proteins were loaded onto NuPAGE^{TM} 3–8% Tris-acetate gels for SDS-PAGE and subsequent immunoblotting analyses.

Immunoblotting analysis and antibodies

After SDS-PAGE, proteins were transferred to a 0.2-μm nitrocellulose membrane (GE, #G5678144) for immunoblotting. Pgk1 was used as a loading control. Antibodies used were anti-myc (Thermo Fisher 9E10). anti-Srs2 (Santa Cruz, yC-18), anti-Pgk1 (Invitrogen, 22C5D8), F9 (a kind gift from Marco Foiani and Daniele Piccini, The FIRC Institute of Molecular Oncology, Milan, Italy) (Pellicioli et al., 2001), anti-Rad9 (a kind gift from John Petrini, MSKCC, NY, USA) (Usui, Foster, & Petrini, 2009), anti-Mec1-S1964-p (a kind gift from James E. Haber, Brandeis University, MA, USA) (Memisoglu et al., 2019).

Validation of antibodies is provided either by the manufacturers or in the cited references. Membranes were scanned with a Fujifilm LAS-3000 luminescent image analyzer, which has a linear dynamic range of 10^4. The signal intensities of nonsaturated bands were measured using ImageJ software. For graphs, data are shown as mean and SD. Statistical differences were determined using two-sided Student's t tests.

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Author contributions

All authors were involved in research design and data analyses, J. F., N. D., T. Y., and V. Y. performed research. J. F. and X.Z. wrote the paper with all authors' input.

Competing interest statement

The authors declare no competing interests.

Supplemental Materials

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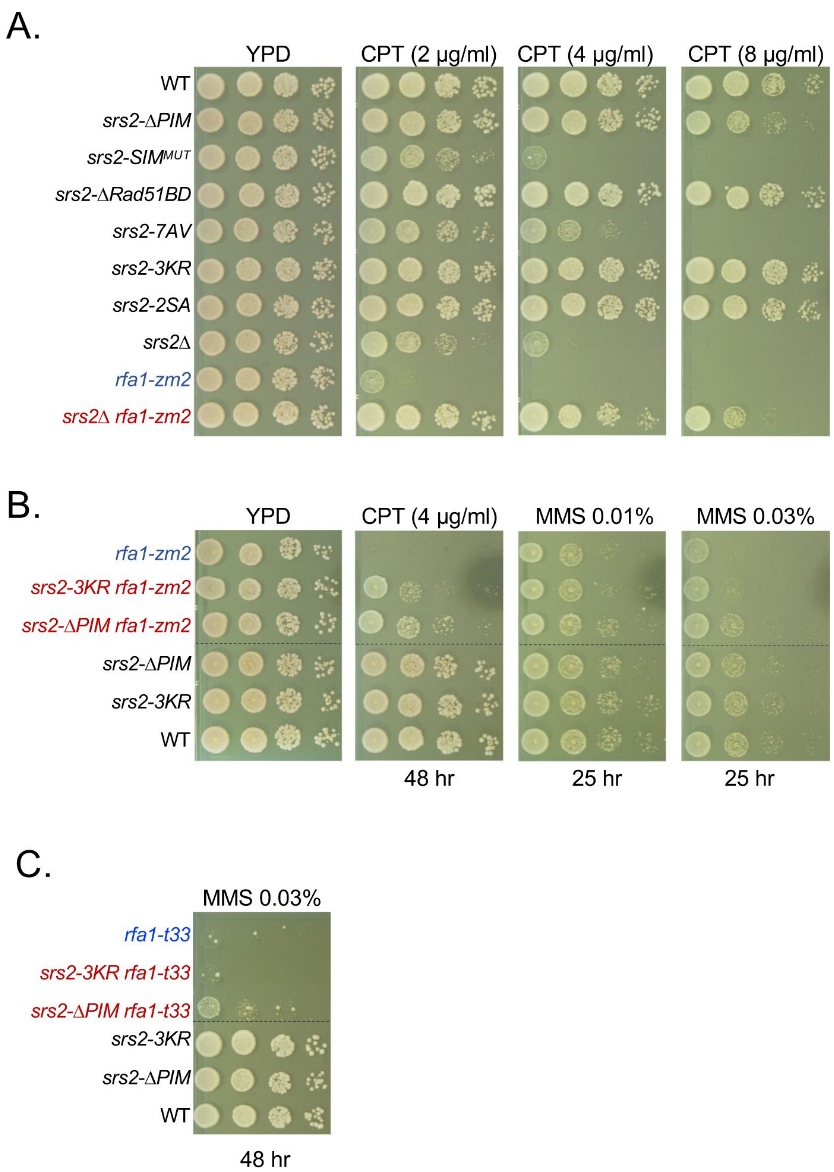

Figure S1.

srs2 mutants examined for CPT sensitivity and interactions with *rfa1* mutant.

(A) *srs2* mutants exhibit different levels of CPT sensitivity. *srs2-ΔPIM* showed slow growth on 8 μg/ml CPT. Sensitivity *srs2-SIM^{MUT}* toward CPT was seen at three CPT concentrations. *srs2-7AV* showed slow growth on media containing 4 and 8 μg/ml CPT. *srs2Δ* and *rfa1-zm2* are mutually suppressive for CPT sensitivity. Experiments are performed as in **Figure 2C**.
(B) *srs2-ΔPIM* confers better suppression of *rfa1-zm2's* CPT and MMS sensitivity than *srs2-3KR*. We note that mild suppression of *rfa1-zm2* by *srs2-3KR* was only seen at 0.03% MMS plates. Experiments are performed as in **Figure 2C**, and incubation times are noted. Dashed lines indicate the removal of superfluous rows.
(C) *srs2-ΔPIM* suppresses *rfa1-t33* sensitivity toward MMS. Dashed lines indicate the removal of superfluous rows.

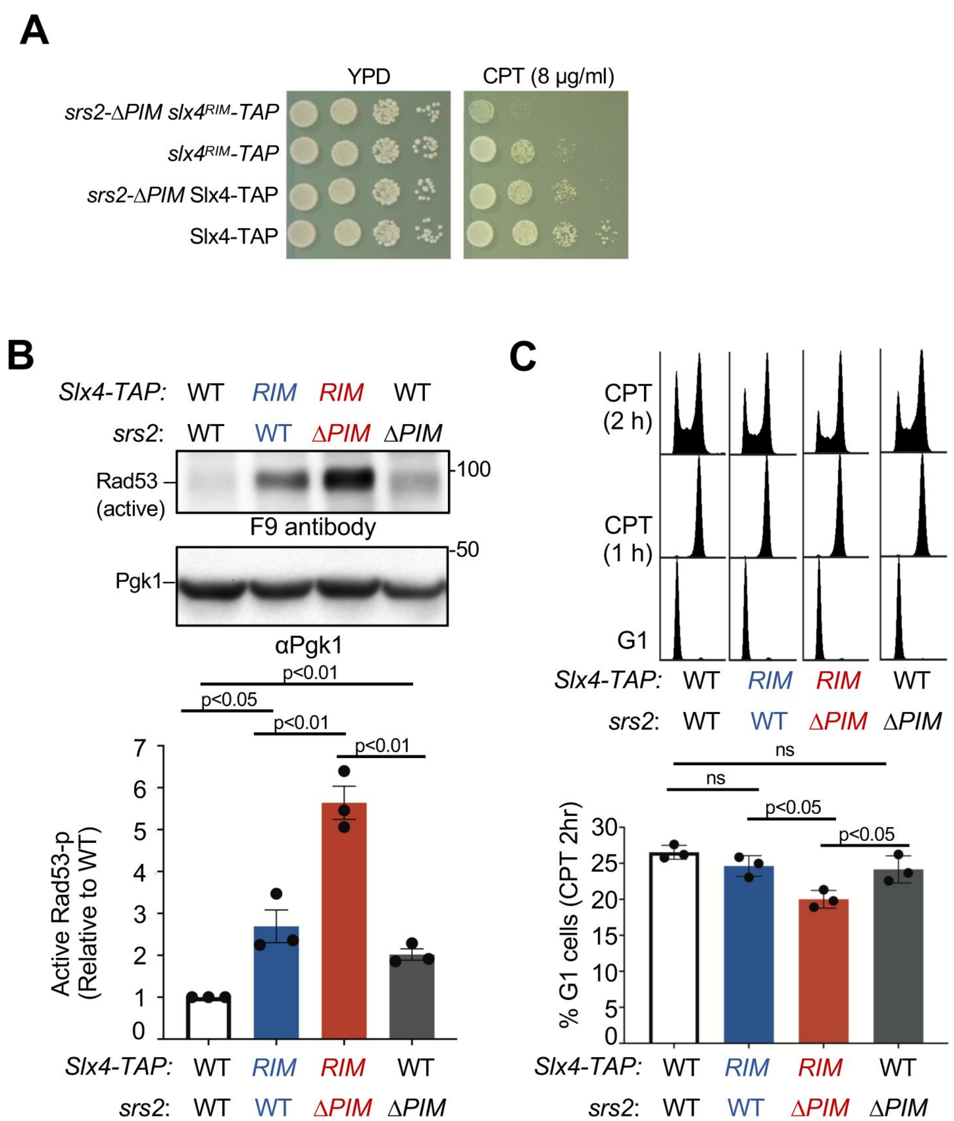

Figure S2.

srs2-ΔPIM and *slx4*^{RIM} are additively defective in DDC down-regulation.

(A) The *srs2-ΔPIMΔ slx4*^{RIM} double mutant shows stronger CPT sensitivity than either single mutant. Experiments are performed as in **Figure 2C**.

(B) *srs2-ΔPIM* and *slx4*^{RIM} are additive for increasing the level of active Rad53 when cells are treated with CPT. Experiments were conducted and quantifications are presented as those described in **Figure 3A**, except that three biological isolates are included.

(C) *srs2-ΔPIM* and *slx4*^{RIM} are additive for reducing cells exiting from G2/M arrest at 2 hours post CPT treatment. Experiments were conducted and quantifications are presented as those described in **Figure 3B**, except that three biological isolates are included.

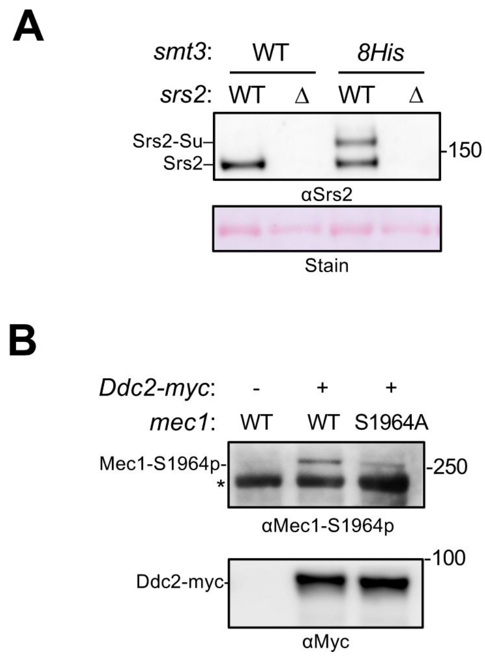

Figure S3.

Confirmation of Srs2 sumoylation and Mec1-S1964 phosphorylation.

(A) Detection of Srs2 sumoylation. Protein extracts from asynchronous cells were bound to Ni-NTA beads and the elutes were examined by immunoblotting using an anti-Srs2 antibody. Sumoylated Srs2 form was detected in wild-type cells expressing 8His-tagged SUMO (Smt3) but not in cells lacking this construct or in *srs2Δ* cells. Unmodified Srs2 showed unspecific Ni-NTA bead binding, thus being detectable in cells containing Srs2 regardless of the SUMO status.
(B) Detection of Mec1-S1964 phosphorylation. Cells were treated with CPT for 2 hours before immunoprecipitating Ddc2-myc was conducted. Ddc2 and the co-immunoprecipitated Mec1 were examined by immunoblotting. Phosphorylation of Ser1964 of the Mec1 protein was detected using the anti-Mec1-S1964-p antibody. This antibody also detected a non-specific band (*). Cells containing untagged Ddc2 or the *mec1-S1964A* mutation were used as controls.

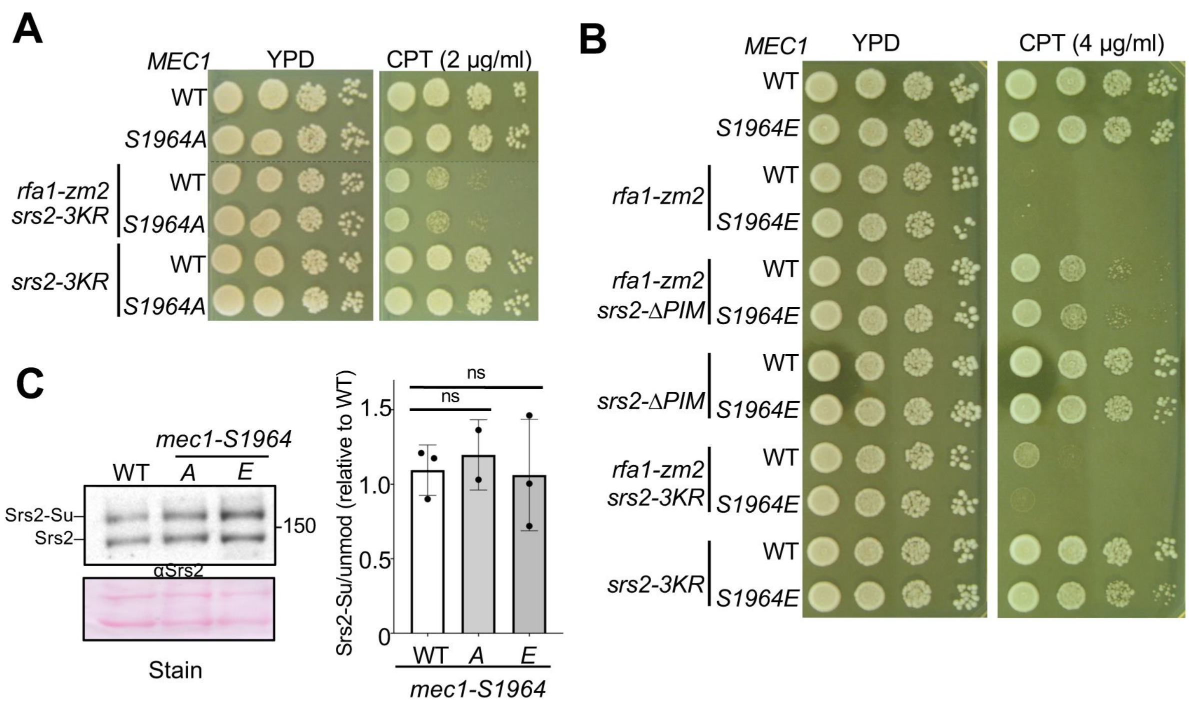

Figure S4.

Examination of Mec1-S964 phosphorylation.

(A) *mec1-S1964A* does not affect the CPT sensitivity of *srs2-3KR* or *srs2-3KR rfa1-zm2* cells. Experiments were performed as in **Figure 2C**.

(B) The effects of *mec1-S1964E* on the CPT sensitivity of *srs2-3KR* and *srs2-ΔPIM* cells with or without *rfa1-zm2*. Experiments were performed as in **Figure 2C**.

(C) *mec1-S1964A* or *-S1964E* does not affect Srs2 sumoylation. Experiments were performed as in **Figure 4A**. Mean values of three biological isolates per genotype are graphed with error bars representing SD. ns, not significant.

Table S1

Strains used in this study.

All strains are derived from W1588-4C, a *RAD5* derivative of W303 (*MATa ade2-1 can1-100 ura3-1 his3-11,15, leu2-3, 112 trp1-1 rad5-535*). Only one strain is listed per each genotype, but at least two independent isolates of each genotype were used in the experiments.

Strain	Genotype	Source
X8665-12-22D	*MATa srs2-ΔPIM*	This study
X8198-15B	*srs2-SIM*^{MUT}	This study
G1114	*MATa srs2-7AV::HIS3*	(Saponaro et al., 2010)
X8650-2-1D	*MATa srs2-3KR::NAT*	This study
X8659-14A	*MATa srs2-2SA*	This study
G854	*MATα srs2-Δ(875-902)::HPH*	(Colavito et al., 2009)
X8047-1B	*MATα rfa1-zm2*	(Dhingra et al., 2021)
X8650-2C	*MATa srs2-3KR::NAT rfa1-zm2*	This study
X8666-35-6A	*MATa srs2-ΔPIM rfa1-zm2*	This study
X8210-2B	*MATa rfa1-t33*	(Dhingra et al., 2021)
X8745-2A	*MATa srs2-3KR::NAT rfa1-t33*	This study
X8743-13A	*MATα srs2-ΔPIM rfa1-t33*	This study
X8198-15A	*srs2-SIM*^{MUT} *rfa1-zm2*	This study
X8637-6B	*MATa srs2-7AV::HIS3 rfa1-zm2*	This study
X8137-10B	*MATa srs2-Δ875-902::HPH rfa1-zm2*	This study
X8660-4B	*MATα srs2-2SA rfa1-zm2*	This study
T1358-2	*MATa 8His-SMT3::TRP1*	Lab Collection
X8808-2D	*MATa srs2-ΔPIM 8His-SMT3::TRP1*	This study
X8795-9D	*MATa srs2-3KR::NAT 8His-SMT3::TRP1*	This study
X8807-6A	*MATa sml1Δ::URA3 8His-SMT3::TRP1*	This study
X8746-4D	*MATa sml1Δ::URA3 mec1Δ::TRP1 8His-SMT3::TRP1*	This study
X8807-7A	*MATa sml1Δ::URA3 rad53Δ::HIS3 8His-SMT3::TRP1*	This study
G461	*MATa 3FLAG-RAD53::LEU2*	J. Petrini
X9055-1D	*MATa DDC2-Myc18::HIS3*	This study
X9225-3A	*MATa mec1-S1964A*	This study
X9225-3C	*MATa mec1-S1964A rfa1-zm2*	This study
X9088-38C	*MATa mec1-S1964A srs2-ΔPIM*	This study
X9225-1B	*MATα mec1-S1964A srs2-ΔPIM rfa1-zm2*	This study
X8495-5D	*MATa rfa1-S178D*	This study
X8495-2D	*MATa srs2Δ::HIS3 rfa1-S178D*	This study
X8494-3B	*MATa rfa1-S178A*	This study
X8494-3A	*MATa srs2Δ::HIS3 rfa1-S178A*	This study
X8658-14-22D	*MATa rfa2-3SA*	This study
X8658-14-13C	*MATa srs2Δ::HIS3 rfa2-3SA*	This study
X8779-13D	*MATa rfa2-3SD*	This study
X8779-13B	*MATa srs2Δ rfa2-3SD*	This study
X9265-4D	*MATa SLX4-TAP::HIS3*	(Wan et al., 2019)
X9265-4A	*MATa srs2-ΔPIM SLX4-TAP::HIS3*	This study
X8956-7C	*MATa slx4*^{RIM}*-TAP::HIS3*	(Wan et al., 2019)
X8956-5D	*MATa srs2-ΔPIM slx4*^{RIM}*-TAP::HIS3*	This study
X9228-2D	*MATa mec1-S1964A srs2-3KR::NAT*	This study
X9228-5A	*MATα mec1-S1964A srs2-3KR::NAT rfa1-zm2*	This study
X9090-9C	*MATa mec1-S1964E*	This study
X9091-14A	*MATa mec1-S1964E rfa1-zm2*	This study
X9090-19A	*MATa mec1-S1964E srs2-ΔPIM*	This study
X9132-15B	*MATa mec1-S1964E srs2-ΔPIM rfa1-zm2*	This study
X9091-13A	*MATa mec1-S1964E srs2-3KR::NAT*	This study
X9132-10B	*MATα mec1-S1964A srs2-3KR::NAT rfa1-zm2*	This study
X9137-3B	*MATa mec1-S1964A 8His-SMT3::TRP1*	This study
X9136-2B	*MATa mec1-S1964E 8His-SMT3::TRP1*	This study

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Editors

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Reviewer #1 (Public Review):

Overall, the data presented in this manuscript is of good quality. Understanding how cells control RPA loading on ssDNA is crucial to understanding DNA damage responses and genome maintenance mechanisms. The authors used genetic approaches to show that disrupting PCNA binding and SUMOylation of Srs2 can rescue the CPT sensitivity of rfa1 mutants with reduced affinity for ssDNA. In addition, the authors find that SUMOylation of Srs2 depends on binding to PCNA and the presence of Mec1.

Noted weaknesses include the lack of evidence supporting that Srs2 binding to PCNA and its SUMOylation occur at ssDNA gaps, as proposed by the authors. Also, the mutants of Srs2 with impaired binding to PCNA or impaired SUMOylation showed no clear defects in checkpoint dampening, and in some contexts, even resulted in decreased Rad53 activation. Therefore, key parts of the paper would benefit from further experimentation and/or clarification.

Major Comments

(1) The central model proposed by the authors relies on the loading of PCNA at the 3' junction of an ssDNA gap, which then mediates Srs2 recruitment and RPA removal. While several aspects of the model are consistent with the data, the evidence that it is occurring at ssDNA gaps is not strong. The experiments mainly used CPT, which generates mostly DSBs. The few experiments using MMS, which mostly generates ssDNA gaps, show that Srs2 mutants lead to weaker rescue in this context (Figure S1). How do the authors explain this discrepancy? In

the context of DSBs, are the authors proposing that Srs2 is engaging at later steps of HR-driven DSB repair where PCNA gets loaded to promote fill-in synthesis? If so, is RPA removal at that step important for checkpoint dampening? These issues need to be addressed and the final model adjusted.

(2) The data in Figure 3 showing that Srs2 mutants reduce Rad53 activation in the rfa1-zm2 mutant are confusing, especially given the claim of an anti-checkpoint function for Srs2 (in which case Srs2 mutants should result in increased Rad53 activation). The authors propose that Rad53 is hyperactivated in rfa1-zm2 mutant because of compromised ssDNA protection and consequential DNA lesions, however, the effects sharply contrast with the central model. Are the authors proposing that in the rfa1-zm2 mutant, the compromised protection of ssDNA supersedes the checkpoint-dampening effect? Perhaps a schematic should be included in Figure 3 to depict these complexities and help the reader. The schematic could also include the compensatory dampening mechanisms like Slx4 (on that note, why not move Figure S2 to a main figure?... and even expand experiments to better characterize the compensatory mechanisms, which seem important to help understand the lack of checkpoint dampening effect in the Srs2 mutants)

(3) The authors should demarcate the region used for quantifying the G1 population in Figure 3B and explain the following discrepancy: By inspection of the cell cycle graph, all mutants have lower G1 peak height compared to WT (CPT 2h). However, in the quantification bar graph at the bottom, ΔPIM has higher G1 population than the WT.

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Reviewer #2 (Public Review):

Summary:

This is an interesting paper that delves into the post-translational modifications of the yeast Srs2 helicase and proteins with which it interacts in coping with DNA damage. The authors use mutants in some interaction domains with RPA and Srs2 to argue for a model in which there is a balance between RPA binding to ssDNA and Srs2's removal of RPA. The idea that a checkpoint is being regulated is based on observing Rad53 and Rad9 phosphorylation (so there are the attributes of a checkpoint), but evidence of cell cycle arrest is lacking. The only apparent delay in the cell cycle is the re-entry into the second S phase (but it could be an exit from G2/M); but in any case, the wild-type cells enter the next cell cycle most rapidly. No direct measurement of RPA residence is presented.

Strengths:

Data concern viability assays in the presence of camptothecin and in the post-translational modifications of Srs2 and other proteins.

Weaknesses:

There are a couple of overriding questions about the results, which appear technically excellent. Clearly, there is an Srs2-dependent repair process here, in the presence of camptothecin, but is it a consequence of replication fork stalling or chromosome breakage? Is repair Rad51-dependent, and if so, is Srs2 displacing RPA or removing Rad51 or both? If RPA is removed quickly what takes its place, and will the removal of RPA result in lower DDC1-MEC1 signaling?

Moreover, It is worth noting that in single-strand annealing, which is ostensibly Rad51 independent, a defect in completing repair and assuring viability is Srs2-dependent, but this defect is suppressed by deleting Rad51. Does deleting Rad51 have an effect here?

Neither this paper nor the preceding one makes clear what really is the consequence of having a weaker-binding Rfa1 mutant. Is DSB repair altered? Neither CPT nor MMS are necessarily good substitutes for some true DSB assay.

With camptothecin, in the absence of site-specific damage, it is difficult to test these questions directly. (Perhaps there is a way to assess the total amount of RPA bound, but ongoing replication may obscure such a measurement). It should be possible to assess how CPT treatment in various genetic backgrounds affects the duration of Mec1/Rad53-dependent checkpoint arrest, but more than a FACS profile would be required.

It is also notable that MMS treatment does not seem to yield similar results (Fig. S1).

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Reviewer #3 (Public Review):

The superfamily I 3'-5' DNA helicase Srs2 is well known for its role as an anti-recombinase, stripping Rad51 from ssDNA, as well as an anti-crossover factor, dissociating extended D-loops and favoring non-crossover outcome during recombination. In addition, Srs2 plays a key role in ribonucleotide excision repair. Besides DNA repair defects, srs2 mutants also show a reduced recovery after DNA damage that is related to its role in downregulating the DNA damage signaling or checkpoint response. Recent work from the Zhao laboratory (PMID: 33602817) identified a role of Srs2 in downregulating the DNA damage signaling response by removing RPA from ssDNA. This manuscript reports further mechanistic insights into the signaling downregulation function of Srs2.

Using the genetic interaction with mutations in RPA1, mainly rfa1-zm2, the authors test a panel of mutations in Srs2 that affect CDK sites (srs2-7AV), potential Mec1 sites (srs2-2SA), known sumoylation sites (srs2-3KR), Rad51 binding (delta 875-902), PCNA interaction (delta 1159-1163), and SUMO interaction (srs2-SIMmut). All mutants were generated by genomic replacement and the expression level of the mutant proteins was found to be unchanged. This alleviates some concern about the use of deletion mutants compared to point mutations. The double mutant analysis identified that PCNA interaction and SUMO sites were required for the Srs2 checkpoint dampening function, at least in the context of the rfa1-zm2 mutant. There was no effect of these mutants in a RFA1 wild-type background. This latter result is likely explained by the activity of the parallel pathway of checkpoint dampening mediated by Slx4, and genetic data with an Slx4 point mutation affecting Rtt107 interaction and checkpoint downregulation support this notion. Further analysis of Srs2 sumoylation showed that Srs2 sumoylation depended on PCNA interaction, suggesting sequential events of Srs2 recruitment by PCNA and subsequent sumoylation. Kinetic analysis showed that sumoylation peaks after maximal Mec1 induction by DNA damage (using the Top1 poison camptothecin (CPT)) and depended on Mec1. These data are consistent with a model that Mec1 hyperactivation is ultimately leading to signaling downregulation by Srs2 through Srs2 sumoylation. Mec1-S1964 phosphorylation, a marker for Mec1 hyperactivation and a site found to be needed for checkpoint downregulation after DSB induction did not appear to be involved in checkpoint downregulation after CPT damage. The data are in support of the model that Mec1 hyperactivation when targeted to RPA-covered ssDNA by its Ddc2 (human ATRIP) targeting factor, favors Srs2 sumoylation after Srs2 recruitment to PCNA to disrupt the RPA-Ddc2-Mec1 signaling complex. Presumably, this allows gap filling and disappearance of long-lived ssDNA as the initiator of checkpoint signaling, although the study does not extend to this step.

Strengths

(1) The manuscript focuses on the novel function of Srs2 to downregulate the DNA damage signaling response and provide new mechanistic insights.

(2) The conclusions that PCNA interaction and ensuing Srs2-sumoylation are involved in checkpoint downregulation are well supported by the data.

Weaknesses

(1) Additional mutants of interest could have been tested, such as the recently reported Pin mutant, srs2-Y775A (PMID: 38065943), and the Rad51 interaction point mutant, srs2-F891A (PMID: 31142613).

(2) The use of deletion mutants for PCNA and RAD51 interaction is inferior to using specific point mutants, as done for the SUMO interaction and the sites for post-translational modifications.

(3) Figure 4D and Figure 5A report data with standard deviations, which is unusual for n=2. Maybe the individual data points could be plotted with a color for each independent experiment to allow the reader to evaluate the reproducibility of the results.

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Author response:

> ***eLife assessment***:
>
> *This manuscript reports valuable findings on the role of the Srs2 protein in turning off the DNA damage signaling response initiated by Mec1 (human ATR) kinase. The data provide solid evidence that Srs2 interaction with PCNA and ensuing SUMO modification is required for checkpoint downregulation. However, experimental evidence with regard to the model that Srs2 acts at gaps after camptothecin-induced DNA damage is currently lacking. The work will be of interest to cell biologists studying genome integrity but would be strengthened by considering the possible role of Rad51 and its removal.*

We appreciate the editors and the reviewers for providing evaluation and helpful comments. As detailed below, we plan to adjust the writing and figures to address the points raised by the reviewers. We believe that these changes will improve the clarity of the work. Below is a summary of our plan to address the two main criticisms.

> *(1) Regarding the sites of Srs2 action, our data support the conclusion that Srs2 removal of RPA is favored at a subset of ssDNA regions that have proximal PCNA, but not at sites lacking PCNA. A logical supposition for the former types of ssDNA regions includes ssDNA gaps and tails generated during DNA repair or replication, wherein PCNA can be loaded at the ssDNA-dsDNA junction with a 3' DNA end. Examples of the latter type of ssDNA regions without proximal PCNA can form within negatively supercoiling regions or intact R-loops, both of which lack 3' DNA end for PCNA loading. While we have stated this conclusion in the text, we highlighted ssDNA gaps as sites of Srs2 action in Discussion and in the model figure, which could be misleading. We will clarify our model, that is, Srs2 distinguishes among different types of ssDNA regions using PCNA proximity as a guide for RPA removal, and state that the precise nature of Srs2 action sites remain to be determined. Regardless, the feature of Srs2 revealed in this work provides a rationale for how it can remove RPA at subsets of ssDNA regions without unnecessary stripping of RPA at other sites.*

(2) While Rad51 removal is an important facet of Srs2 functions, it is not relevant to our current study based on the following observations and rationales.

First, we have provided several lines of evidence to support the conclusion that Rad51 removal by Srs2 is separable from the Srs2-RPA antagonism (Dhingra et al.*, 2021). For example, while* rad51Δ *rescues the hyper-recombination phenotype of* srs2Δ *cells, it does not affect the hyper-checkpoint phenotype of* srs2Δ*. Strikingly,* rfa1-zm1/zm2 *have the opposite effect. The differential effects of* rad51Δ *and* rfa1-zm1/zm2 *were also seen for the _srs2-_ATPase dead allele (*srs2-K41A*). For example,* rfa1-zm2 *rescued the hyper-checkpoint defect and the CPT sensitivity of* srs2-K41A*, while* rad51Δ *had neither effect.*

These and other data described in Dhingra et al *suggest that Srs2's effects on checkpoint vs. recombination are separable and that the Srs2-RPA antagonism during the DNA damage checkpoint is independent of Rad51.*

Second, our current work addresses which Srs2 features affect the Srs2-RPA antagonism during the DNA damage response and its implications. Given this antagonism is separable from Srs2 removal of Rad51, including Rad51 regulation would be distractive from the main points of this work.

*Third, in the current work, we began by examining all known regulatory and protein-protein interaction features of Srs2, including the Rad51 binding domain. Consistent with our conclusion summarized above based on the Dhingra et al study, deleting the Rad51 binding domain in Srs2 (*srs2-ΔRad51BD) *has no effect on* rfa1-zm2 *phenotype in CPT (Figure 2D). This is in sharp contrast to mutating the PCNA binding and the sumoylation sites of Srs2, which suppressed* rfa1-zm2 *for its CPT sensitivity and checkpoint abnormalities (Figure 2C). This data provides yet another evidence that Srs2 regulation of Rad51 is separable from the Srs2-RPA antagonism.*

In summary, our work provides a foundation for future examination of how Srs2 regulates RPA and Rad51 in different manners, how these two facets of the Srs2 functions affect genome integrity in different capacity, and whether there is a crosstalk between them during certain DNA metabolism processes.

Public Reviews:

Reviewer #1:

Overall, the data presented in this manuscript is of good quality. Understanding how cells control RPA loading on ssDNA is crucial to understanding DNA damage responses and genome maintenance mechanisms. The authors used genetic approaches to show that disrupting PCNA binding and SUMOylation of Srs2 can rescue the CPT sensitivity of rfa1 mutants with reduced affinity for ssDNA. In addition, the authors find that SUMOylation of Srs2 depends on binding to PCNA and the presence of Mec1. Noted weaknesses include the lack of evidence supporting that Srs2 binding to PCNA and its SUMOylation occur at ssDNA gaps, as proposed by the authors. Also, the mutants of Srs2 with impaired binding to PCNA or impaired SUMOylation showed no clear defects in checkpoint dampening, and in some contexts, even resulted in decreased Rad53 activation. Therefore, key parts of the paper would benefit from further experimentation and/or clarification.

We thank the reviewer for the positive comments on this work and address her/his remark regarding ssDNA gaps below in Major Comment #1. In addition, we detailed below our data and rationale in suggesting that the checkpoint dampening phenotype of *srs2-ΔPIM* and *-3KR* (deficient for PCNA binding and sumoylation, respectively) is masked by redundant pathways. We further describe our plan to enhance the clarity of both text and model to address these points from the reviewer.

> *Major Comments*
>
> *(1) The central model proposed by the authors relies on the loading of PCNA at the 3' junction of an ssDNA gap, which then mediates Srs2 recruitment and RPA removal. While several aspects of the model are consistent with the data, the evidence that it is occurring at ssDNA gaps is not strong. The experiments mainly used CPT, which generates mostly DSBs. The few experiments using MMS, which mostly generates ssDNA gaps, show that Srs2 mutants lead to weaker rescue in this context (Figure S1). How do the authors explain this discrepancy? In the context of DSBs, are the authors proposing that Srs2 is engaging at later steps of HRdriven DSB repair where PCNA gets loaded to promote fill-in synthesis? If so, is RPA removal at that step important for checkpoint dampening? These issues need to be addressed and the final model adjusted.*

We appreciate the reviewer's concern. Our conclusion is that Srs2 can be guided by PCNA to a subset of ssDNA regions for RPA removal, and that this Srs2 action is not favored at ssDNA regions with no proximal PCNA. It is important to note that CPT can produce both types of ssDNA regions. Besides ssDNA generated via DSB-associated recombinational repair, CPT can also lead to ssDNA gap formation upon excision repair and DNA-protein crosslink repair of trapped Top1 (Sun *et al.*, 2020). ssDNA regions generated during these DNA repair processes often contain 3' DNA end for PCNA loading, thus they can favor Srs2 removal of RPA. Another facet of CPT's effects (besides DNA lesions) is depleting functional pool of Top1, thus causing topological stress and consequently increased levels of DNA supercoiling and R-loops (Koster *et al.*, 2007, Petermann *et al.*, 2022). ssDNA formed within the negatively supercoiled regions and in R-loops lacks 3' DNA end unless it is cleaved by nucleases, thus these sites would be disfavored for Srs2 removal of RPA due to lack of PCNA loading. Our conclusion that ssDNA regions with nearby PCNA are preferred sites for Srs2 action provides a rationale for how Srs2 can remove RPA at certain ssDNA regions but minimize unnecessary stripping of RPA from other sites.

We will clarify in Discussion that CPT can generate twp types of ssDNA regions as stated above, and that Srs2 could distinguish among them using PCNA proximity as a guide for RPA removal. While this conclusion was described in the text, we emphasized ssDNA gap as a Srs2 action site in the model. We will clarify that while this is a logical supposition, other types of ssDNAs with proximal PCNA could also be targeted by Srs2 and that our work paves the way to determine the precise nature of ssDNA regions for Srs2's action.

The reasons for the less potent growth suppression of *rfa1* mutants by *srs2* alleles in MMS condition compared with CPT condition are unclear, but multiple possibilities should be considered, given that MMS and CPT affect checkpoint responses differently and that RPA and Srs2 affect growth in multiple ways. For example, while CPT only activates the DNA damage checkpoint, MMS additionally induces DNA replication checkpoint (Menin *et al.*, 2018, Redon *et al.*, 2003). It is thus possible that the Srs2-RPA antagonism is relatively more important for the DNA damage checkpoint than the DNA replication checkpoint. Further investigation of this possibility among others will shed light on differential suppressive effects seen in this work. We will include this discussion in the revised text.

> *(2) The data in Figure 3 showing that Srs2 mutants reduce Rad53 activation in the rfa1-zm2 mutant are confusing, especially given the claim of an anti-checkpoint function for Srs2 (in which case Srs2 mutants should result in increased Rad53 activation). The authors propose that Rad53 is hyperactivated in rfa1-zm2 mutant because of compromised ssDNA protection and consequential DNA lesions, however, the effects sharply contrast with the central model. Are the authors proposing that in the rfa1-zm2*

> *mutant, the compromised protection of ssDNA supersedes the checkpoint-dampening effect? Perhaps a schematic should be included in Figure 3 to depict these complexities and help the reader. The schematic could also include the compensatory dampening mechanisms like Slx4 (on that note, why not move Figure S2 to a main figure?... and even expand experiments to better characterize the compensatory mechanisms, which seem important to help understand the lack of checkpoint dampening effect in the Srs2 mutants)*

Genetic interactions that involve partially defective alleles, multi-functional proteins, and redundant pathways are complex to comprehend. For example, a phenotype seen for the null allele may not be seen for partially defective alleles. In the context of this study, while *srs2* null increased Rad53 activation (Dhingra *et al.*, 2021), *srs2-ΔPIM* and *-3KR* did not (Figure 3A-3B). However, *srs2-ΔPIM* enhanced Rad53 activation when combined with another checkpoint dampening mutant *slx4RIM*, suggesting that defects of *srs2-ΔPIM* can be compensated by Slx4 (Figure S2). Importantly, *srs2-ΔPIM* and *-3KR* rescued *rfa1-zm2*'s checkpoint abnormality (Figure 3A3B), suggesting that Srs2 binding to PCNA and its sumoylation contribute to the Srs2-RPA antagonism in the DNA damage checkpoint response.

A partially defective allele that impairs a specific function of a protein can be a powerful genetic tool even when it lacks a particular phenotype on its own. For example, a partially defective allele of the checkpoint protein Rad9 impairing its binding to gamma-H2A (*rad9-K1088M*) does not affect the G2/M checkpoint nor cause DNA damage sensitivity due to the compensation of other checkpoint factors (Hammet *et al.*, 2007); however_, rad9-K1088M_ rescues the DNA damage sensitivity and persistent G2/M checkpoint of *rtt107* and *slx4* mutants, providing one of the evidences supporting a role of the Slx4-Rtt107 axis in removal of Rad9 from chromatin (via competing with Rad9 for gamma-H2A binding) (Ohouo *et al.*, 2013).

In order to highlight the checkpoint recovery process, the model in Figure 6 did not depict another consequence of the Srs2-RPA antagonism. In the presence of Srs2, DNA binding *rfa1* mutants can lead to increased levels of DNA lesions and checkpoint, and these defects are rescued by lessening Srs2's ability to strip RPA from DNA (Dhingra *et al.*, 2021). We will modify the model in Figure 6 and its legend to clarify that the model depicts just one of the consequences of the Srs2 and RPA antagonism with a focus on the checkpoint recovery. We will also state these points more clearly in the Discussion. Further, a new schematic in Figure 3 as suggested by the reviewer will be added to outline the genetic relationship and interpretation. We will also follow reviewer's suggestion to move Figure S2 to the main figures. Better characterizing the compensatory mechanisms among different checkpoint dampening pathways is very interesting but requires substantial amounts of work. While it is beyond the scope of the current study, it could be pursued in the future.

> *(3) The authors should demarcate the region used for quantifying the G1 population in Figure 3B and explain the following discrepancy: By inspection of the cell cycle graph, all mutants have lower G1 peak height compared to WT (CPT 2h). However, in the quantification bar graph at the bottom, ΔPIM has higher G1 population than the WT.*

We have added the description on how the G1 region of the FACS histogram was selected to derive the percentage of G1 cells in Figure 3B. Briefly, for samples collected for a particular strain, the G1 region of the "G1 sample" was used to demarcate the G1 region of the "CPT 2h" sample. Upon re-checking the included FACS profiles, we realized that a mutant panel and its datapoint were mistakenly put in the place for wild-type. We will correct this mistake. The conclusion remains that *srs2-ΔPIM* and *srs2-3KR* improved *rfa1-zm2* cells' ability to exit G2/M, while they themselves do not show difference from the wild-type control for the percentage of G1 cells after 2hr CPT treatment. We will add statistics in figures to reflect this conclusion and adjust the order of strains shown in panel A and B to be consistent with each other.

> ***Reviewer #2:***
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> *This is an interesting paper that delves into the post-translational modifications of the yeast Srs2 helicase and proteins with which it interacts in coping with DNA damage. The authors use mutants in some interaction domains with RPA and Srs2 to argue for a model in which there is a balance between RPA binding to ssDNA and Srs2's removal of RPA. The idea that a checkpoint is being regulated is based on observing Rad53 and Rad9 phosphorylation (so there are the attributes of a checkpoint), but evidence of cell cycle arrest is lacking. The only apparent delay in the cell cycle is the re-entry into the second S phase (but it could be an exit from G2/M); but in any case, the wild-type cells enter the next cell cycle most rapidly. No direct measurement of RPA residence is presented.*

We thank the reviewer for the helpful comments. Previous studies have shown that CPT does not induce the DNA replication checkpoint, thus it does not slow down or arrest S phase progression; however, CPT does induce the DNA damage checkpoint, which causes a delay of G2/M cells to re-enter into the second cell cycle (Menin *et al.*, 2018, Redon *et al.*, 2003). Our result is consistent with previous findings, showing that CPT induces G2/M delay but not arrest. We will adjust the text to make this point clearer.

We have previously reported chromatin-bound RPA levels in *rfa1-zm2*, *srs2*, and their double mutants, as well as in vitro ssDNA binding by wild-type and mutant RPA complexes (Dhingra *et al.*, 2021). We found that Srs2 loss or its ATPase dead mutant led to 4-6 fold increase of RPA levels on chromatin, which was rescued by *rfa1-zm2* (Dhingra *et al.*, 2021). On its own, *rfa1-zm2* did not cause defective chromatin association in our assays, despite modestly reducing ssDNA binding in *vitro* (Dhingra *et al.*, 2021). This discrepancy could be due to a lack of sensitivity of chromatin fractionation assay in revealing moderate changes of RPA residence on DNA. Considering this, we decided to employ functional assays (Figure 2-3) that are more effective in identifying the Srs2 features pertaining to RPA regulation.

> *Strengths:*
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> *Data concern viability assays in the presence of camptothecin and in the post-translational modifications of Srs2 and other proteins.*
>
> *Weaknesses:*
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> *There are a couple of overriding questions about the results, which appear technically excellent. Clearly, there is an Srs2-dependent repair process here, in the presence of camptothecin, but is it a consequence of replication fork stalling or chromosome breakage? Is repair Rad51-dependent, and if so, is Srs2 displacing RPA or removing Rad51 or both? If RPA is removed quickly what takes its place, and will the removal of RPA result in lower DDC1-MEC1 signaling?*

While Srs2 can affect both the checkpoint response and DNA repair in CPT conditions, the *rfa1-zm2* allele, which affects the former but not the latter, role of Srs2, allows us to gain a deeper understanding of the former role (Dhingra *et al.*, 2021). This role also appears to be critical for cell survival in CPT, since *srs2*Δ growth on CPT-containing media was greatly improved by *rfa1-zm* mutants (Dhingra *et al.*, 2021). Building on this understanding, our current study identified two Srs2 features that could afford spatial and temporal regulations of RPA removal from DNA, thus providing a rationale for how cells can properly utilize this beneficial yet also dangerous activity. Study of Srs2-mediated repair in CPT conditions, either in Rad51-dependent or independent manner, before and after replication forks stall or DNA breaks, will require substantial efforts and can be pursued in the future. We will add this point to the revised manuscript.

> *Moreover, it is worth noting that in single-strand annealing, which is ostensibly Rad51 independent, a defect in completing repair and assuring viability is Srs2-dependent, but this defect is suppressed by deleting Rad51. Does deleting Rad51 have an effect here?*

We have shown in our previous paper (Dhingra *et al.*, 2021). that *rad51Δ* did not rescue the hyper-checkpoint phenotype of *srs2Δ* cells in CPT condition (Dhingra *et al.*, 2021), while *rfa1-zm1* and *-zm2* did (Dhingra *et al.*, 2021). Such differential effects were also seen for the *srs2* ATPase-dead allele (Dhingra *et al.*, 2021). These and other data described in the Dhingra et al paper suggest that Srs2's effects on checkpoint vs. recombination are separable at least in CPT condition, and that the Srs2-RPA antagonism in checkpoint regulation is not affected by Rad51 removal (unlike in SSA situation).

> *Neither this paper nor the preceding one makes clear what really is the consequence of having a weakerbinding Rfa1 mutant. Is DSB repair altered? Neither CPT nor MMS are necessarily good substitutes for some true DSB assay.*

In our previous report (Dhingra *et al.*, 2021), we showed that the *rfa1-zm* mutants did not affect the frequencies of rDNA recombination, gene conversation, or direct repeat repair (Dhingra *et al.*, 2021). Further, *rfa1-zm* mutants did not suppress the hyper-recombination phenotype of *srs2Δ*, while *rad51Δ* did (Dhingra *et al.*, 2021). In a DSB system, wherein the direct repeats flanking the break were placed 30 kb away from each other, *srs2Δ* led to hyper-checkpoint and lethality, both of which were rescued by *rfa1-zm* mutants (Dhingra *et al.*, 2021). In this assay, *rfa1-zm* mutants themselves did not show sensitivity, suggesting the repair is largely proficient. Collectively, these data provide evidence to suggest that weaker DNA binding of Rfa1 does not have detectable effect on the recombinational repair assays examined thus far, rather it has a profound effect in Srs2-mediated checkpoint downregulation. In-depth studies of *rfa1-zm* mutations in the context of various DSB repair steps will be interesting to pursue in the future.

> *With camptothecin, in the absence of site-specific damage, it is difficult to test these questions directly. (Perhaps there is a way to assess the total amount of RPA bound, but ongoing replication may obscure such a measurement). It should be possible to assess how CPT treatment in various genetic backgrounds affects the duration of Mec1/Rad53-dependent checkpoint arrest, but more than a FACS profile would be required.*

Quantitative measurement of RPA residence time on DNA in cells and the duration of Mec1/Rad53-dependent checkpoint arrest will be very informative but requires further technology development. Our current work provides a foundation for such quantitative assessment.

> *It is also notable that MMS treatment does not seem to yield similar results (Fig. S1).*

Figure S1 showed that *srs2-ΔPIM* and *srs2-3KR* had weaker suppression of *rfa1-zm2* growth on MMS plates than on CPT plates. The reasons for the less potent growth suppression in MMS condition compared with CPT condition are unclear, but multiple possibilities should be considered, given that MMS and CPT affect checkpoint responses differently and that RPA and Srs2 affect growth in multiple ways. For example, while CPT only activates the DNA damage checkpoint, MMS additionally induces DNA replication checkpoint (Menin *et al.*, 2018, Redon *et al.*, 2003). It is thus possible that the Srs2-RPA antagonism is more important for the DNA damage checkpoint than the DNA replication checkpoint. Further investigation of this and other possibilities will provide clues to the differential suppressive effects seen in this work. We will include this discussion in the revised text.

> ***Reviewer #3:***
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> *The superfamily I 3'-5' DNA helicase Srs2 is well known for its role as an anti-recombinase, stripping Rad51 from ssDNA, as well as an anti-crossover factor, dissociating extended D-loops and favoring non-crossover outcome during recombination. In addition, Srs2 plays a key role in ribonucleotide excision repair. Besides DNA repair defects, srs2 mutants also show a reduced recovery after DNA damage that is related to its role in downregulating the DNA damage signaling or checkpoint response. Recent work from the Zhao laboratory (PMID: 33602817) identified a role of Srs2 in downregulating the DNA damage signaling response by removing RPA from ssDNA. This manuscript reports further mechanistic insights into the signaling downregulation function of Srs2.*
>
> *Using the genetic interaction with mutations in RPA1, mainly rfa1-zm2, the authors test a panel of mutations in Srs2 that affect CDK sites (srs2-7AV), potential Mec1 sites (srs2-2SA), known sumoylation sites (srs2-3KR), Rad51 binding (delta 875-902), PCNA interaction (delta 1159-1163), and SUMO interaction (srs2SIMmut). All mutants were generated by genomic replacement and the expression level of the mutant proteins was found to be unchanged. This alleviates some concern about the use of deletion mutants compared to point mutations. The double mutant analysis identified that PCNA interaction and SUMO sites were required for the Srs2 checkpoint dampening function, at least in the context of the rfa1-zm2 mutant. There was no effect of these mutants in a RFA1 wild-type background. This latter result is likely explained by the activity of the parallel pathway of checkpoint dampening mediated by Slx4, and genetic data with an Slx4 point mutation affecting Rtt107 interaction and checkpoint downregulation support this notion. Further analysis of Srs2 sumoylation showed that Srs2 sumoylation depended on PCNA interaction, suggesting sequential events of Srs2 recruitment by PCNA and subsequent sumoylation. Kinetic analysis showed that sumoylation peaks after maximal Mec1 induction by DNA damage (using the Top1 poison camptothecin (CPT)) and depended on Mec1. These data are consistent with a model that Mec1 hyperactivation is ultimately leading to signaling downregulation by Srs2 through Srs2 sumoylation. Mec1-S1964 phosphorylation, a marker for Mec1 hyperactivation and a site found to be needed for checkpoint downregulation after DSB induction did not appear to be involved in checkpoint downregulation after CPT damage. The data are in support of the model that Mec1 hyperactivation when targeted to RPA-covered ssDNA by its Ddc2 (human ATRIP) targeting factor, favors Srs2 sumoylation after Srs2 recruitment to PCNA to disrupt the RPA-Ddc2-Mec1 signaling complex. Presumably, this allows gap filling and disappearance of long-lived ssDNA as the initiator of checkpoint signaling, although the study does not extend to this step.*
>
> *Strengths*
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> *(1) The manuscript focuses on the novel function of Srs2 to downregulate the DNA damage signaling response and provide new mechanistic insights.*
>
> *(2) The conclusions that PCNA interaction and ensuing Srs2-sumoylation are involved in checkpoint downregulation are well supported by the data.*

We thank the reviewer for carefully reading our work and for his/her positive comments.

> *Weaknesses*
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> *(1) Additional mutants of interest could have been tested, such as the recently reported Pin mutant, srs2Y775A (PMID: 38065943), and the Rad51 interaction point mutant, srs2-F891A (PMID: 31142613).*

srs2-Y775A was shown to be proficient for stripping RPA from ssDNA and behaved like wild-type Srs2 in assays such as gene conversion and crossover control, and exhibited a genetic interaction profile as the wildtype allele. The authors suggest that the Y775 pin can contribute to unwinding secondary DNA structures. Collectively, these findings do not provide a strong rationale for *srs2-Y775A* being relevant for RPA removal from ssDNA.

We have already included the data showing that a *srs2* mutant lacking the Rad51 binding domain (*srs2-ΔRad51BD*, Δ875-902) did not affect *rfa1-zm2* growth in CPT nor caused other defects in CPT on its own (Figure 2D). This data suggest that Rad51 binding is not relevant to the Srs2-RPA antagonism in CPT, a conclusion fully supported by data in our previous study (Dhingra *et al.*, 2021). Collectively, these findings do not provide a strong rationale to test a point mutation within the Rad51BD region.

> *(2) The use of deletion mutants for PCNA and RAD51 interaction is inferior to using specific point mutants, as done for the SUMO interaction and the sites for post-translational modifications.*

We agree with this view generally. However, this is less of a concern for the Rad51 binding site mutant (*srs2ΔRad51BD*), as it behaved as the wild-type allele in our assays. The *srs2-ΔPIM* mutant (lacking 4 amino acids) has been examined for PCNA binding in vitro and in vivo in several studies (e.g. Kolesar *et al.*, 2016, Kolesar *et al.*, 2012); to our knowledge no unintended defect was reported. We thus believe that this allele is suitable for testing whether Srs2's ability to bind PCNA is relevant to RPA regulation.

> *(3) Figure 4D and Figure 5A report data with standard deviations, which is unusual for n=2. Maybe the individual data points could be plotted with a color for each independent experiment to allow the reader to evaluate the reproducibility of the results.*

We will include individual data points as suggested and correct figure legend to indicate that three independent biological samples per genotype were examined in both panels.

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